“A phase 1, multicenter, open-label, prospective, first-in-human dose-escalation clinical trial of Domain Therapeutics’ anti-CCR8 monoclonal antibody (DT-7012) in patients with relapsed or refractory cutaneous T-cell lymphomas (CTCL)”
CITY

CLINICAL TRIAL ON MEDICINAL PRODUCT FOR HUMAN USE

Version N°1.0 dated 02/06/2025
Project Code: APHP240605 / EU CT number: 2024-520294-13-00

Coordinating investigator	Prof Adele DE MASSON Dermatology Saint-Louis Hospital, Paris Tel. +33171207501/adele.demasson@aphp.fr
Scientific Director	Prof Marie BEYLOT-BARRY Dermatology Saint André Hospital, CHU Bordeaux Tel. +33557822500/marie.beylot-barry@chu-bordeaux.fr
Sponsor	Assistance Publique – Hôpitaux de Paris (AP-HP) and by delegation: Direction de la Recherche Clinique et de l’Innovation - DRCI (Clinical Research and Innovation Department) Hôpital Saint-Louis 1, avenue Claude Vellefaux 75010 PARIS DRCI-Head Office project advisor: Christine Pereira / Marine Camus Tel. +33140271857/Email christine.pereira@aphp.fr Tel. +33140274076/Email marine.camus2@aphp.fr
Entity responsible for monitoring the trial	Unité de Recherche Clinique (URC) Adresse Hôpital Saint-Louis, 1 av Vellefaux 75010 Paris Référent projet URC : Emmanuelle SACCO Tél +33142385394 / Email emmanuelle.sacco@aphp.fr
Pharmaceutical coordination	Agence Générale des Equipements et Produits de Santé (AGEPS) Département Essais Cliniques 7 Rue du Fer à moulin 75005 PARIS Tel. +33146691402
Entity responsible for Safety	Safety department – DRCI Hôpital Fernand Widal 200 rue du Faubourg Saint-Denis 75475 Paris Sarra DALIBEY, head of safety department Tel. +33140275785/Email expertisecsi.drc@aphp.fr

Direction de la Recherche Clinique et de l’Innovation - DRCI (Clinical Research and Innovation Department)
Hôpital Saint Louis 75010 PARIS

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

SIGNATURE page for a research PROTOCOL

Research code number: APHP240605 / EU CT n°2024-520294-13-00

Title: “A phase 1, multicenter, open-label, prospective, first-in-human dose-escalation clinical trial of Domain Therapeutics’ anti-CCR8 monoclonal antibody (DT-7012) in patients with relapsed or refractory cutaneous T-cell lymphomas (CTCL)” - CITY

Version no 1.0 dated: 02/06/2025

The study will be carried out in accordance with the protocol, with current good practices and with statutory and regulatory requirements.

Coordinating Investigator:

Pr Adèle DE MASSON
Department of Dermatology
Hospital Saint-Louis
Paris

Date:
Signature:

Sponsor

Assistance Publique – Hôpitaux de Paris
Direction de la Recherche Clinique et de l’Innovation - DRCI (Clinical Research and Innovation Department)
Hôpital Saint Louis
1 avenue Claude Vellefaux
75010 PARIS

Date:
Signature:

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

TABLE OF CONTENTS

1. SUMMARY 8

2. SCIENTIFIC JUSTIFICATION FOR THE STUDY 12

2.1. HYPOTHESIS FOR THE STUDY 12
2.2. DESCRIPTION OF KNOWLEDGE RELATING TO THE CONDITION INVOLVED 12
2.3. SUMMARY OF RELEVANT PRE-CLINICAL EXPERIMENTS AND CLINICAL TRIALS 13
2.4. DESCRIPTION OF THE POPULATION TO BE STUDIED AND JUSTIFICATION FOR THIS CHOICE OF PARTICIPANTS 15
2.5. IDENTIFICATION AND DESCRIPTION OF THE INVESTIGATIONAL MEDICATION OR MEDICATIONS 15
2.6. DESCRIPTION AND JUSTIFICATION OF THE DOSAGE, ROUTE OF ADMINISTRATION, ADMINISTRATION SCHEDULE AND TREATMENT DURATION 15
2.7. SUMMARY OF THE KNOWN AND FORESEEABLE BENEFITS AND RISKS FOR THE CLINICAL TRIAL PARTICIPANTS 15

3. OBJECTIVES 16

3.1. PRIMARY AND SECONDARY OBJECTIVES 16
3.2. OBJECTIVE OF ANY POTENTIAL EXPLORATORY STUDY 17

4. STUDY DESIGN 17

4.1. STUDY ENDPOINTS 17
4.1.1. Primary endpoint 17
4.1.2. Secondary endpoints 17
4.1.3. Exploratory endpoints 18
4.2. DESCRIPTION OF RESEARCH METHODOLOGY 18
4.2.1. Design of the study 18
4.2.2. Number of participating sites 19
4.2.3. Identification of participants 20

5. IMPLEMENTATION OF THE STUDY 20

5.1. SCREENING VISIT 20
5.2. BASELINE VISIT/INCLUSION VISIT 21
5.3. TREATMENT VISITS 22
5.4. FOLLOW-UP ± TREATMENT VISITS 22
5.5. VISIT FOR SUSPECTED PROGRESSIVE DISEASE 23
5.6. LAST STUDY VISIT 23
5.7. EARLY TERMINATION VISIT 23
5.8. EXPECTED LENGTH OF PARTICIPATION AND DESCRIPTION OF THE CHRONOLOGY AND DURATION OF THE STUDY 24
5.9. TABLE OR DIAGRAM SUMMARIZING THE CHRONOLOGY OF THE STUDY 24
5.10. DISTINCTION BETWEEN STANDARD CARE AND STUDY 25
5.11. BIOLOGICAL SAMPLES COLLECTION – FOR RESEARCH 26

6. ELIGIBILITY CRITERIA 29

6.1. INCLUSION CRITERIA 29
6.2. EXCLUSION CRITERIA 30
6.3. CONTRACEPTION DURING THE STUDY 31
6.3.1. Definition of women of childbearing potential and of fertile men 31
6.3.2. Pregnancy control requirements 31
6.3.3. Birth control methods which may be considered as highly effective 31
6.4. RECRUITMENT PROCEDURE 31
6.5. TERMINATION RULES 32
6.5.1. Termination rules for individual participants 32
6.5.2. Procedures for replacing participants if applicable 34
6.5.3. Termination rules for the clinical trial: full or partial discontinuation of the clinical trial 34

7. TREATMENT ADMINISTERED TO STUDY PARTICIPANTS 34

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

7.1. DESCRIPTION OF THE INVESTIGATIONAL MEDICINAL PRODUCT(S) 34
7.1.1. Investigational medicinal product *34*
7.2. DESCRIPTION OF TRACEABILITY ELEMENTS ACCOMPANYING THE INVESTIGATIONAL MEDICINAL PRODUCT(S) 35
7.3. DESCRIPTION OF AUXILIARY MEDICINAL PRODUCT(S) (TREATMENTS REQUIRED TO CONDUCT THE STUDY) 36
7.4. AUTHORISED AND PROHIBITED TREATMENTS (MEDICINAL, ADDITIONAL MEDICINAL, SURGICAL), INCLUDING RESCUE MEDICATIONS 36
7.4.1. Prohibited medications *36*
7.4.2. Permitted medications *36*
7.5. METHODS FOR MONITORING COMPLIANCE WITH THE TREATMENT 36
7.6. DESCRIPTION OF THE TREATMENT MODALITIES 37

8. EFFICACY ASSESSMENT 37

8.1. EVALUATION OF EFFICACY 37
8.1.1. Response criteria ..... ***Erreur ! Signet non défini.***
8.1.2. Response in Skin *37*
8.1.3. Response in Nodes *37*
8.1.4. Response in Viscera *38*
8.1.5. Response in Blood *39*
8.1.6. Global Response Score *39*
8.2. ENDPOINT DEFINITION 40
8.2.1. Overall Response rate *40*
8.2.2. Disease control rate *40*
8.2.3. Progression Free Survival Rate *40*
8.2.4. Duration of response *41*
8.2.5. Time to next treatment *41*

9. SPECIFIC STUDY COMMITTEES 41

9.1. STEERING COMMITTEE 41
9.2. SCIENTIFIC COMMITTEE 41

10. SAFETY ASSESSMENT - RISKS AND BURDEN ADDED BY THE STUDY 41

10.1. DESCRIPTION OF SAFETY ENDPOINTS ASSESSMENT PARAMETERS 41
10.2. ANTICIPATED METHODS AND TIMETABLE FOR MEASURING, COLLECTING AND ANALYSING THE SAFETY ENDPOINTS 42
10.3. RECORDING AND REPORTING ADVERSE EVENTS 42
10.3.1. Definitions *42*
10.3.2. The role of the investigator *42*
10.3.3. Role of the sponsor *46*
10.3.4. Data Safety Monitoring Board (DSMB) *47*

11. DATA MANAGEMENT 48

11.1. DATA COLLECTION PROCEDURES 48
11.2. IDENTIFICATION OF DATA RECORDED DIRECTLY IN THE CRFS WHICH WILL BE CONSIDERED AS SOURCE DATA 48
11.3. RIGHT TO ACCESS DATA AND SOURCE DOCUMENTS 48
11.3.1. Data access *48*
11.3.2. Source documents *48*
11.3.3. Data confidentiality *48*
11.4. DATA PROCESSING AND STORAGE OF RESEARCH DOCUMENTS AND DATA 48
11.4.1. Identification of the data processing manager and location(s) *48*
11.4.2. Data entry *49*
11.5. DATA OWNERSHIP 49

12. STATISTICAL ASPECTS 49

12.1. DESCRIPTION OF STATISTICAL METHODS TO BE USED INCLUDING THE TIMETABLE FOR THE PLANNED INTERIM ANALYSES 49
12.2. CALCULATION HYPOTHESES FOR THE NUMBER OF PARTICIPANTS REQUIRED AND THE RESULTS 50
12.3. ANTICIPATED LEVEL OF STATISTICAL SIGNIFICANCE 50

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

12.4. STATISTICAL CRITERIA FOR TERMINATION OF THE STUDY. 50
12.5. METHOD FOR TAKING INTO ACCOUNT MISSING, UNUSED OR INVALID DATA 50
12.6. CHOICE OF INDIVIDUALS TO BE INCLUDED IN THE ANALYSES 51

13. QUALITY CONTROL AND ASSURANCE 51

13.1. GENERAL ORGANISATION 51
13.1.1. Strategy for centre opening 51
13.1.2. Scope of centre monitoring 51
13.2. QUALITY CONTROL 51
13.3. CASE REPORT FORMS 52
13.4. MANAGEMENT OF NON-COMPLIANCES 52
13.5. AUDITS/INSPECTIONS 52
13.6. PRINCIPAL INVESTIGATOR'S COMMITMENT TO ASSUME RESPONSIBILITY 52
13.7. SUITABILITY OF THE FACILITIES 53

14. ETHICAL AND LEGAL CONSIDERATIONS 53

14.1. METHODS FOR INFORMING RESEARCH PARTICIPANTS AND OBTAINING THEIR CONSENT 53
14.2. PROHIBITION FROM PARTICIPATING IN ANOTHER CLINICAL STUDY OR EXCLUSION PERIOD SET AFTER THE STUDY 53
14.3. COMPENSATION FOR PARTICIPANTS 54
14.4. REGISTRATION ON THE NATIONAL REGISTER OF STUDY PARTICIPANTS TO CLINICAL TRIAL CONCERNING THE PRODUCTS MENTIONNED IN ARTICLE L. 5311-1 OF THE CODE DE LA SANTÉ PUBLIQUE 54
14.5. AUTHORISATION FOR THE RESEARCH LOCATION 54
14.6. LEGAL OBLIGATIONS 54
14.6.1. Role of the sponsor 54
14.6.2. Request for authorisation 54
14.6.3. Procedures relating to data protection regulations 54
14.6.4. Start of the Clinical Trial 55
14.6.5. Amendments to the Clinical Trial 55
14.6.6. End of the Clinical Trial 55
14.6.7. Summary of the results of the clinical trial 55
14.6.8. Archiving 55

15. FUNDING AND INSURANCE 55

15.1. FUNDING SOURCES 56
15.2. INSURANCE 56

16. PUBLICATION RULES 56

16.1. MENTION OF AP-HP AFFILIATION FOR PROJECTS SPONSORED BY AP-HP 56
16.2. MENTION OF THE SPONSOR AP-HP (DRCI) IN THE ACKNOWLEDGEMENTS OF THE TEXT 56
16.3. MENTION OF THE FINANCIAL BACKER IN THE ACKNOWLEDGEMENTS OF THE TEXT 56

17. BIBLIOGRAPHY 56

19. LIST OF ADDENDA 58

19.1. LIST OF INVESTIGATORS 58
19.2. SERIOUS ADVERSE EVENTS NOTIFICATION FORM 58
19.3. PREGNANCY NOTIFICATION FORM 58
19.4. SMPC OR INVESTIGATOR'S BROCHURE 58
19.5. QUESTIONNAIRE OR SCALE 58
19.6. DESCRIPTION OF THE CLINICAL TRIAL IN THE AP-HP TRIALS REGISTER 63

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

ABBREVIATIONS / GLOSSARY

ADCC antibody-dependent cellular cytotoxicity
AE adverse event
ALP alkaline phosphatase
ALT alanine aminotransferase
AST aspartate aminotransferase
ATL adult T-cell leukemia-lymphoma
CCR4 C-C chemokine receptor type 4
CD cluster of differentiation antigen
cfDNA cell-free circulating tumor DNA
CHO Chinese hamster ovary
CIF confirmation of interest by Principal Investigator Form
CL clearance
CR complete response
CRF case report form
CTCAE common terminology criteria for adverse events
CTCL cutaneous T-cell lymphoma
CV% % coefficient of variation
DLT dose-limiting toxicity
ECG electrocardiogram
EGA European Genome-Phenome Archive
EMA European Medicines Agency
EORTC European Organization for Research and Treatment of Cancer
EORTC-QLQ-C30 European Organization for Research and Treatment of Cancer Quality of Life Questionnaire
FMF folliculotropic mycosis fungoides
GRS global response score
HBM human biological material
HBsAb hepatitis B core antibody
HBsAg hepatitis B surface antigen
HBV hepatitis B virus
HIV human immunodeficiency virus
HR hazard ratio
IB Investigator Brochure
ICH/GCP International Council for Harmonisation/Good Clinical Practice
Ig immunoglobulin
IgG1κ immunoglobulin G subclass 1 kappa
IL interleukin
INN International Nonproprietary Name
IQR interquartile range
ISCL International Society for Cutaneous Lymphomas
ITT Intent-to-treat
kD kilodalton
LVEF left ventricular ejection fraction
mAb monoclonal antibody
MDC macrophage-derived chemokine
MF Mycosis fungoides
mSWAT modified severity weighted assessment tool
NCI National Cancer Institute
ORR objective response rate
OS overall survival
PBG Pathobiology Group
PD progressive disease
PFS progression-free survival

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

PFSR-48 progression free survival rate at 48 weeks
PK pharmacokinetics
PR partial response
PTCL peripheral T-cell lymphoma
ROC Receiver operating characteristic
SAE serious adverse events
SD stable disease
SEER Surveillance, Epidemiology and End Results
SJS Stevens-Johnson syndrome
SmPC summary of product characteristics
SS Sézary syndrome
T3 triiodothyronine
T4 thyroxine
TEAE treatment-emergent adverse event
TR translational research
Treg T regulatory cells
TSH thyroid stimulating hormone
ULN upper limit of normal
USCLC United States Cutaneous Lymphoma Consortium
VAS: Visual Analogue Scale
WHO World Health Organization
WOCBP WOMEN of childbearing potential

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

1. SUMMARY

Full title	A phase 1, multicenter, open-label, prospective, first-in-human dose-escalation clinical trial of Domain Therapeutics' anti-CCR8 monoclonal antibody (DT-7012) in patients with relapsed or refractory cutaneous T-cell lymphomas (CTCL)
Acronym/reference	CITY (CCR8 In cutaneous T cell lYmphomas)
Coordinating investigator	Pr Adèle de Masson
Scientific Director	Pr Marie Beylot-Barry
Methodologist	Dr Lucie Biard
Sponsor	Assistance Publique – Hôpitaux de Paris
Scientific justification	Cutaneous T-cell lymphomas (CTCL) are a heterogeneous group of lymphomas characterized by a primary involvement of the skin. Among them, mycosis fungoides (MF) and Sézary syndrome (SS) are the most common subtypes. SS is defined as erythroderma (erythema of the entire skin surface), and circulating tumor blood cells. The circulating tumor T cells express CD4 and may lose expression of CD7 and CD26, while exhibiting in most cases aberrant expression of CD158k (KIR3DL2), which is a surface marker of Sézary cells. CCR8 is a surface marker of tumor-infiltrating regulatory T cells. We have recently observed that CCR8 was expressed by tumor cells in CTCL and other peripheral T-cell lymphomas. CCR8 is expressed by skin resident-memory T cells which are believed to be the tumor cell-of-origin in mycosis fungoides. Domain Therapeutics (DT) showed the in vitro efficacy of their proprietary anti-CCR8 mAb DT-7012 in the depletion of CTCL cells. Therapeutic depletion of CCR8-expressing cells by DT-7012 could eliminate tumor cells and activate the anti-tumor immunity in CTCL. We hypothesize that treatment with DT-7012 is effective in the treatment of relapsed or refractory (R/R) CTCL as advanced MF and SS.
Main objective and primary endpoint	In patients with R/R CTCL treated with single agent DT-7012: • Primary Objective o To identify the maximum tolerated dose (MTD) of intravenous dosing of single agent DT-7012 in terms of short-term safety & tolerability • Primary endpoint Dose limiting toxicity (DLT) defined by any grade ≥3 toxicity (CTCAE v5.0) lasting >7 days within 3 weeks from the first DT-7012 administration, except for: o in patients without lymphopenia at inclusion: lymphopenia considered as DLT only if ≥ grade 4 o in patients with lymphopenia at inclusion: more than 80% of reduction of lymphocytes count which is considered DLT, o skin AE and infectious AE (very frequent in CTCL patients) considered as DLT only if ≥ grade 4 (life-threatening).
Secondary objectives and endpoints	• Secondary Objectives 1. To assess DT-7012 safety (adverse events) 2. To assess DT-7012 efficacy(response) 3. To assess pharmacokinetics of DT-7012 4. To evaluate the immunogenicity of DT-7012 5. To assess DT-7012 effect on pruritus (patient reported outcome) 6. To assess health-related quality of life (patient reported outcome) 7. To assess Time To Next Treatment

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

	8. To assess disease control rate (DCR: ORR + stable disease) 9. To assess duration of response (DoR) 10. To assess progression-free survival (PFS) • Secondary Endpoints 1. Safety endpoints: Adverse Events (AEs), Serious Adverse Events (SAEs), Drug related AEs, Drug related SAEs, Adverse Events of Special Interest (AESI) up to 12 months 2. Efficacy endpoint: objective response rate (ORR, complete response (CR) + partial response (PR) at 3 months, Response assessment will be performed according to the Global Response Score of the International Society for Cutaneous Lymphomas/European Organization of Research and Treatment of Cancer (ISCL/EORTC) criteria for MF/SS (Olsen, et al. 2022). 3. PK endpoints (Maximum and trough concentration of DT-7012 at each administration, Area under the curve from time 0 to day 7 (AUC day 0-7) for the first and fourth administration, Accumulation index in terms of ratio of concentrations and AUC between the fourth and first administration), 4. Immunogenicity endpoint: presence of anti-drug antibodies (ADA) 5. Pruritus assessment, by a visual analogue scale (VAS) at 3, 6 and 12 months 6. Quality of life (EORTC-QLQ C30 and Skindex-29) at 3, 6 and 12 months 7. Time to next treatment (TTNT) up to 12 months 8. Disease control rate (defined as complete, partial response or stable disease), at 3, 6 and 12 months 9. Duration of response (DoR) defined as the time from measurement of CR/PR (whichever is first recorded) until the date of documented recurrent or progressive disease, up to 12 months (Assessed in responder patients only) 10. Progression-free survival (PFS) at 12 months • Exploratory objectives 1. To investigate the association of response with intensity of CCR8 expression in blood and skin at baseline and during follow-up, CCR8 expression will be studied by immunohistochemistry and immunofluorescence in lesional skin at baseline and 3 months, and by flow cytometry in peripheral blood at baseline, 1 and 3 months after inclusion 2. To investigate the immune response to DT-7012 in blood and skin, biomarkers (some markers among, but not limited to exhaustion markers such as e.g., PD1, TIGIT, Tregs markers CD25 and FoxP3, activation markers such as CD69, CD25 and cell markers – CD68, CD163 macrophages, CD8 T cells) will be studied by multiplex immunofluorescence in lesional skin at baseline and 3 months, and biomarkers by flow cytometry in peripheral blood at baseline, 1 and 3 months after inclusion 3. To assess the molecular residual disease by high throughput sequencing of T-cell receptor genes and PCR in skin and blood (cell free and cellular DNA) at 3 months compared to baseline
Design of the study	Single arm, open-label, multicenter prospective phase 1 dose-escalation study using the time-to-event continual reassessment method (TITE-CRM) design [O'Quigley 1990; Cheung 2000]
Category	**Category 2: Phase 1 (risk D)**
Population of study participants	Adult patients with relapsed or refractory CTCL after signed informed consent

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Inclusion criteria	To be eligible for study entry patients must meet all of the following criteria 1. Adult Patients (≥18 years) with no upper age limit 2. Confirmed diagnosis of mycosis fungoides or Sezary syndrome 3. Stage IB to IVB in the ISCL / EORTC classification 4. Relapsed or refractory (no response) after at least two systemic treatments 5. ECOG performance status 0-1 6. Adequate liver function: • Total bilirubin ≤ 1.5 xULN, or Direct bilirubin ≤ 1.5xULN if total bilirubin is >1.5xULN, or total bilirubin >1.5 xULN if elevated total bilirubin is attributed to Gilbert's syndrome or to histologically-proven liver involvement by CTCL • Aspartate aminotransferase (AST) and alanine aminotransferase (ALT) ≤ 2,5 x ULN, unless elevated to up to 5 x ULN due to CTCL 7. Adequate hematological function: • Absolute neutrophil count of ≥ 1.5 G/L without G-CSF support for at least 7 days • Platelet count of ≥ 75 G/L without platelet transfusion within 7 days • Hemoglobin ≥ 9 g/dL without RBC transfusion within 7 days 8. Adequate renal function: creatinine clearance calculated by Cockcroft & Gault formula of ≥ 50 mL/min 9. HBV: negative blood HBs Ag or blood HBV DNA. Vaccinated patients may be included. Patients with HBc antibody may be included if HBV DNA is negative 10. HCV: negative HCV serology, or negative HCV RNA if HCV serology is positive 11. HIV: negative HIV serology 12. Negative serum or urinary pregnancy test within 7 days or at baseline prior to study treatment in women of childbearing potential 13. Patients must agree to use a highly effective contraceptive method from inclusion until: • If the patient is a male: at least 6 months after the last dose of DT-7012. Men must refrain from donating sperm during this same period • If patient is a female of childbearing potential: at least 6 months after the last dose of DT-7012 14. Patients must have the following minimum wash-out from previous treatments: • 12 weeks for total skin electron beam irradiation, • 4 weeks for monoclonal antibodies • 3 weeks for local radiation therapy, systemic cytotoxic anticancer therapy, treatment with other anti-neoplastic investigational agents • 3 weeks for systemic retinoids, interferons, vorinostat, romidepsin, fusion proteins • 3 weeks for phototherapy • 2 weeks for topical therapy (including steroids, retinoids, nitrogen mustard or imiquimod). Topical steroids and oral steroids (10 mg prednisone equivalent/day maximum) are allowed, if the patient has been on a stable dose with stable symptoms for at least 4 weeks prior to study entry. 15. Patient covered by any social security system (registered or being a beneficiary of such a scheme) 16. Signed informed consent

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Exclusion criteria	Patients meeting any of the following criteria will not be included in the study: 1. Known central nervous system involvement by CTCL 2. Participation in any study of a health product within 30 days prior to study entry 3. Patients with a history of other malignancies during the past three years (*except for: non-melanoma skin cancer, lymphomatoid papulosis, curatively treated localized prostate cancer, curatively treated localized breast cancer, resected thyroid cancer, biopsy proven cervical intraepithelial neoplasia or cervical carcinoma in situ, which are not considered exclusion criteria).* 4. Uncontrolled congestive heart failure (New York Heart Association Classification 3 or 4), or angina, myocardial infarction, cerebrovascular accident, transient ischemic attack within 1 month prior to study entry 5. Any severe acute or chronic medical or psychiatric condition 6. Patients with immunodeficiency 7. Any active systemic viral, bacterial, or fungal infection requiring systemic antibiotics within 1 week prior to first study drug dose 8. Chronic use of systemic corticosteroids of prednisone or equivalent >10 mg prednisone equivalent/day for a chronic condition (washout of 8 days from start of treatment is accepted) 9. Other immunosuppressive therapies are also excluded, (washout of 7 days from start of treatment is accepted) 10. Autologous HSCT within 100 days prior to DT-7012 infusion 11. Prior allogeneic HSCT 12. Prior solid organ transplantation 13. Patient with history of confirmed progressive multifocal leukoencephalopathy 14. Known or suspected allergies, hypersensitivity, or intolerance to DT-7012 or its excipients 15. Pregnant or breast-feeding woman, or desire (for both man and woman participant) to conceive a child within 6 months after end of treatment 16. Patient under guardianship or curatorship and protected adults or unable to consent 17. Coagulation disorder contra indicating intravenous infusion 18. History of anaphylactic reaction following vaccination or immunotherapy
Investigational medicinal product(s)	Phase 1 dose-escalation study of DT-7012 given by intravenous infusion QW x5, then Q2W X10 injections, pursued Q4W if clinical benefit
Comparator treatment	None
Interventions added for the study	• DT-7012 given by intravenous infusion Q1W x5, then Q2W x10 injections, pursued Q4W if clinical benefit • CT scan or PET CT at baseline, and at 3-6-9-12 months only **in patients with N3/M1 stage at baseline** • One FFPE skin biopsy at baseline (+ optional frozen skin biopsy at baseline stored on site) • One FFPE skin biopsy at 3 months (+ optional frozen skin biopsy at baseline stored on site) • Tape stripping at baseline, M1 and M3 • Quality of life questionnaires, pruritus VAS • Blood samples before and after each infusion for pharmacokinetics and research
Expected benefits for the participants and for society	We expect a higher disease control of the lymphoma in the participants, compared to standard of care, with a favorable safety profile. As such, we expect an improved quality of life in participants. We also expect a better understanding of the disease pathophysiology.
Practical implementation	This study will use the Bayesian one-stage time-to-event continual reassessment method (TITE-CRM) design for dose finding phase I clinical trials (O'Quigley 1990; Cheung 2000), using an empirical dose-toxicity model with linear weights. A maximum total of 30 patients with CTCL, given 4 candidate dose levels (0.3; 1.0; 3.0; 10.0 mg/kg) will be dose-assigned starting from 1mg/kg dose level, in cohorts of 1 patient and including safety rules notably to ensure staggered accrual.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Number of participants included	Number of participants declared to the insurer (who have signed a consent form): 30
Number of centres	European international study conducted in the dermatology departments of 5 university hospital centers in France, Germany and Spain
Duration of the study	- Enrolment period: 18 months - Participation period (treatment + follow-up): 13 months - Total duration: 31 months
Number of enrolments expected per site and per month	0.3 (on average given the total required sample size and the planned inclusion period)
Statistical analysis	A maximal sample size of 30 patients ensures relevant operating characteristics for the CRM design (probability of correct selection, probability of early stopping, trial duration) in various toxicity scenarios compatible with pre-existing knowledge of the drug class, and considering a maximum tolerable DLT risk of 30% at week 3 [Cheung YK. Clin Trials. 2013;10(6):852-861].
Funding sources	ANR (RHU SPRINT)
Study will have a Data Safety Monitoring Board	Yes

2. SCIENTIFIC JUSTIFICATION FOR THE STUDY

2.1. Hypothesis for the study

We hypothesize that treatment with DT-7012 is safe and effective in the treatment of relapsed or refractory cutaneous T-cell lymphomas.

2.2. Description of knowledge relating to the condition involved

Cutaneous T-cell lymphomas (CTCLs) are a heterogeneous group of rare non-Hodgkin lymphomas characterized by the initial localisation of malignant T-lymphocytes to the skin. CTCLs represent ~75% to 80% of all primary cutaneous lymphomas, according to the current World Health Organization - European Organization for Research and Treatment of Cancer (WHO-EORTC) consensus classification.[1] Mycosis fungoides (MF) is the most common type accounting for 60% of CTCLs. Sezary syndrome (SS) is a rare leukemic form traditionally defined by the triad of pruritic erythroderma, generalized lymphadenopathy and Sezary cell colonization of skin, and peripheral blood.

MF typically affects patients with a median age at diagnosis of 55 to 60 years and a peak in the 70-80 age group, but can be observed in younger population, including children.

Whilst early stage of MF are indolent diseases with long survival (5-year overall survival, 49–100%), advanced MF and SS typically have an aggressive clinical course (<5 years of overall survival in median) [2] . Clinical presentation is usually characterized by patches and plaques in early stages, and tumorsand erythroderma in more advanced stages. Eventually, disease may progress into lymph nodes, peripheral blood or viscera.

Diagnosis of CTCL is based on a combination of clinical, histological, immunophenotypical and occasionally molecular data. MF/SS are classified into 4 distinct stages based on the TNMB (tumor–node–metastasis–blood) classification[3]. MF stages IA to IIA are usually referred to as early stages and stage IIB to IVB as advanced stages of the disease.

According to the EORTC recommendations[4] current standard therapies are:

- For IA-IIA: skin-directed therapies i.e., mechlorethamine, UVB, psoralen plus ultraviolet light therapy (PUVA), topical corticosteroids and localized radiotherapy. Second-line recommended

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

treatments are: retinoids, interferon, total skin electron beam therapy (TSEB), low dose methotrexate, frequently combined with skin-directed therapy.

- For Stage IIB-IIIB: retinoids, interferon, TSEB, monochemotherapy, radiotherapy and mogamulizumab, brentuximab vedotin, romidepsin, polychemotherapy with allogeneic stem cell transplantation as second-line treatment.[5]
- For stage IV: chemotherapy, radiotherapy, mogamulizumab, brentuximab vedotin, romidepsin and allogeneic stem cell transplantation should be considered.

There are very few approved targeted treatments today and current treatments heavily rely on multiple lines of chemotherapy with side effects including immunosuppression and there is a high unmet need of new targeted treatments in CTCL.

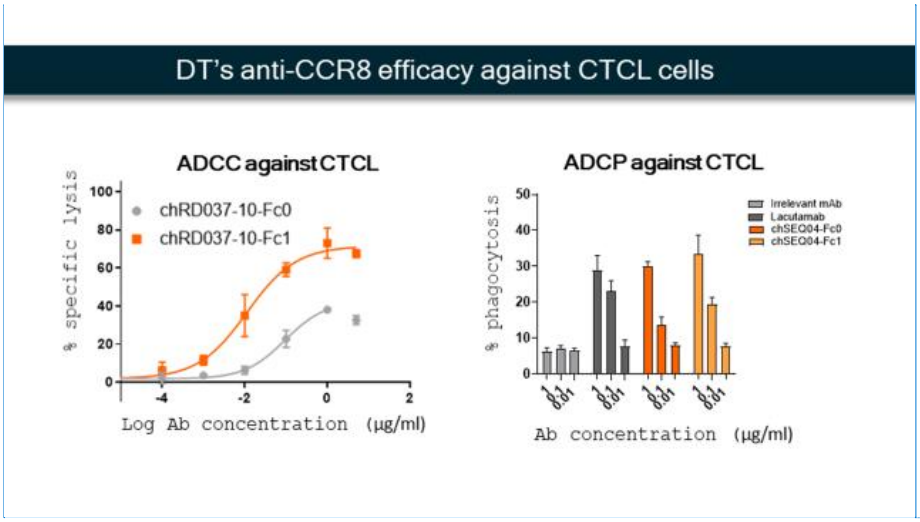
2.3. Summary of relevant pre-clinical experiments and clinical trials

Pre-clinical experiments

We have recently shown that CCR8 was expressed by tumor cells in CTCL and other peripheral T-cell lymphomas such as angioimmunoblastic T-cell lymphomas and hepatosplenic gamma delta T-cell lymphomas.[6]
CCR8 is a chemokine receptor involved in the homing of lymphocytes to skin.[7] CCR8 is expressed by skin resident memory T cells[7] which are suspected to be the tumor cells of origin in MF[8]. CCR8 is also strongly expressed by tumor-infiltrating regulatory T cells (Tregs) involved in immune escape while expression on peripheral blood Tregs is lower. Depletion of CCR8+ Treg exerted a strong antitumor effect independently or in combination with PD-1 inhibitor in tumor mouse models.[9] Anti-CCR8 monoclonal antibody depleted fewer peripheral Tregs than anti-CCR4 monoclonal antibody, suggesting that treatment with therapeutic anti-CCR8 antibody may avoid immune side effects associated with mogamulizumab while sparing antitumor immunity. Therapeutic depletion of CCR8-expressing cells could thus eliminate tumor cells and activate the antitumor immunity in T-cell lymphomas while avoiding immune side effects.

We observed that CCR8 engagement by its ligands CCL18 and CCL1 induced a significant Erk1/2 phosphorylation in tumor cells of CTCL patients. In CTCL patients, PFS was shorter in patients with high expression of CCR8 messenger RNA (hazard ratio, 3.4 (95% confidence interval (CI), 1.6-7.2, P < .001) and CCL18 messenger RNA in skin (hazard ratio, 2.4, 95% CI, 1.5-3.9, P < .0001).[10] At Domain Therapeutics, a drug discovery program aiming at identifying and developing a therapeutic depleting antibody was launched in 2019. Domain has named a drug candidate DT-7012 in 2023 which has been tested against CTCL cell line and showed strong cell reactivity, ADCC and ADCP efficacies (see figure below).

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019



Commenté [CM(1]: sera à cacher pour la soumission du doc public

Patients with higher levels of Treg cells exhibit poorer clinical outcomes and prognoses in several cancers.

In addition, CCR8 is expressed by tumor cells in CTCL and contributes to tumor cell activation and survival in vitro.[10] The depletion of intra tumor CCR8 expressing Treg results in a robust anti-tumor response in syngenic animal models.[9]

Clinical trials

Anti-CCR8 monoclonal antibodies other than DT-7012 have already been tested in human[14] and the most frequent adverse events are underlined hereafter.

A phase 1/2 study (NCT05255484) evaluated the anti-CCR8 monoclonal antibody LM-108 as single agent or in combination with pembrolizumab in patients with advanced solid tumors.

This first-in-human, multicenter, open-label, phase 1/2 study enrolled patients aged ≥ 18 years with advanced solid tumors who had progressed on standard therapy, or were intolerable to the available standard therapy, or had no available standard therapy for treatment. For monotherapy, LM-108 was administered intravenously Q3W at dose levels of 0.3 mg/kg, 1 mg/kg, 3 mg/kg, 10 mg/kg, and 20 mg/kg as per traditional 3+3 design. Combination therapy was evaluated at dose level of 10 mg/kg of LM-108 combined with 200 mg pembrolizumab Q3W.

As of July 3, 2023, 24 patients (21 from monotherapy and 3 from combination therapy) were enrolled, no DLT was observed at all dose levels in monotherapy or combination therapy. MTD was not reached. The most frequent adverse events (AEs) related to the study drug were rash (33%, Grade 1-2) and diarrhea (13%, including 1 Grade 3).

Out of 19 patients from monotherapy, 1 patient achieved partial response (PR) and 11 patients achieved stable disease (SD) with a disease control rate of 63%. One patient achieved PR and one patient achieved SD out of 2 evaluable patients from combination group. The two PR patients had duration of response over 6 months.

LM-108 monotherapy or in combination with pembrolizumab showed excellent safety profile and promising anti-tumor activity in patients with advanced solid tumors.[11,14]

DT-7012 is expected to be first tested in a phase 1 dose escalation trial in monotherapy and in combination with anti-PD-1 pembrolizumab in patients with solid cancer, (NCT 06819735). The first dose to be tested as a monotherapy in the solid cancer trial will be 1 mg/kg with a different administration schedule (Q3W as usually performed in immunotherapy of solid cancers whereas the weekly administration schedule in the first month is usual in CTCL trials).

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

2.4. Description of the population to be studied and justification for this choice of participants

We will include adult patients with CTCL (MF and SS), relapsed or refractory after at least two lines of systemic treatments, because this population has a poor prognosis and limited effective treatment options in the current standard of care.

2.5. Identification and description of the investigational medication or medications

DT-7012 is a Fc enhanced IgG1 that recognizes and binds to an epitope located in the extra-membrane portion of CCR8. The C-C motif chemokine receptor 8 (CCR8) is a class A G protein-coupled receptor (GPCR) that is highly enriched and selectively expressed on intratumoral regulatory T (Treg) cells, which act as suppressors of anti-tumor effector T cell responses. DT-7012 acts by inducing ADCC and ADCP on cells that express CCR8 and by blocking the binding of CCL1, the chemokine agonist for CCR8.

It is anticipated that DT-7012 will eliminate (ADCC and ADCP) and impair the function (CCL1) of tumor infiltrating Treg and CTCL cells and unleash or restore a CD8+ based tumor control.
DT-7012 does not cross react with the murine CCR8 but the in vitro preclinical data generated with DT-7012 support that the data generated with other depleting anti-CCR8 can be extended to DT-7012. In addition, the safety data generated with other IgG1 CCR8 depleting drugs also inform the safety profile of DT-7012 and are informative for the definition of the starting dose.

2.6. Description and justification of the dosage, route of administration, administration schedule and treatment duration

In the present CITY trial in R/R CTCL patients, DT-7012 will be administered every week (QW) for 5 injections, then Q2W for 10 injections, pursued Q4W if clinical benefit, ±2 days with candidate dose levels 0.3; 1.0; 3.0; 10 mg/kg. The starting dose is 1mg/kg.

These dose levels have been selected based on a FIH clinical trial of another anti-CCR8 LM-108 in adult patients with solid malignancies.[11,14] (see Investigator Brochure). In this trial, anti-CCR8 mAb was used with doses up to 20 mg/kg without dose-limiting toxicity. The optimal biological dose for available depleting monoclonal antibodies in CTCL is comprised between 1.0 [12] and 10 mg/kg.[13]

DT-7012 will be administered as intravenous (i.v.) infusion. Participants will receive DT-7012 until confirmed progression, unacceptable toxicity, or any criterion for study treatment withdrawal occurs.
In case of infusion-related reactions at any infusion, a premedication regimen of an anti-histamine and a non-NSAID anti-pyretic (i.v. or oral equivalent) will be given to the patient as per local standard practice for subsequent infusions approximately 30 minutes prior to each dose of study drug.

Immediate access to intensive care unit or equivalent environment and appropriate medical therapy (including i.v. epinephrine, corticosteroids, antihistamines, bronchodilators, and oxygen) must be in place for use in the treatment of potential infusion-related reactions. Infusion of DT-7012 will be stopped in case of Grade >2 infusion-related, allergic, or anaphylactic reactions (according to NC-CTCAE v5.0). Following DT-7012 infusions, participants must be observed for 6 hours post infusion for the first infusion for potential infusion-related reactions, then they will be observed for 2 hours after the subsequent infusions. For the first 3 infusions, a phone call at 24h will be made to check on the patient.

The administration Q1W during the first month, then Q2W for X10 and Q4W thereafter is based on the efficiency of such a treatment schedule in previously tested monoclonal antibodies with the same mode of action (ADCC/ADCP) in CTCL: mogamulizumab [12] and lacutamab [13] whereas monoclonal antibodies tested in a Q3W treatment schedule such as cusatuzumab showed disappointing results in CTCL. [14]

2.7. Summary of the known and foreseeable benefits and risks for the Clinical Trial participants

We expect a higher disease control of the lymphoma in the participants, compared to standard of care. As such, we expect an improved quality of life in participants.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

In the phase 1/2 clinical trial of anti-CCR8 LM-108 in adult patients with solid cancers (see Investigator Brochure), the most frequent adverse events (AEs) related to the study drug were rash (33%, Grade 1-2) and diarrhea (13%, including 1 Grade 3); DLT was not reached at doses up to 20 mg/kg.
Out of 19 patients from monotherapy, 1 patient achieved partial response (PR) and 11 patients achieved stable disease (SD) with a disease control rate of 63%. One patient achieved PR and one patient achieved SD out of 2 evaluable patients from combination group. The two PR patients had duration of response over 6 months.
LM-108 monotherapy or in combination with pembrolizumab thus showed excellent safety profile and promising anti-tumor activity in patients with advanced solid tumors.[11,14]

We thus expect a favorable safety profile of DT-7012 which has the same mode of action as LM-108, and a majority of skin and digestive adverse events.
Given:

- the prognosis of advanced stage MF/SS with median overall survival <5 years,
- the limited availability of effective treatments,
- and the possible transfusion support and planned dose adaptation protocol in case of hematological toxicity,

the risk/benefit ratio justifies the conduct of the study.

The risks are associated with the potential adverse events associated with:

- Intravenous infusion medication DT-7012 (see above)
- PET-CT:
    - 1/Risks associated with tracer (^{18}FDG) extravasation. The volume of injected tracer is very small. If extravasation should occur, patients should be informed and monitored to detect potential adverse effects, such as erythema, or burns. These side effects are very rare with radiopharmaceuticals for diagnostic use.
    - 2/Risks associated with FDG injection. No reaction to FDG injection has been described. However, any allergic reaction (pruritus, rash…) to the injection should be reported and documented. Prior history of reaction to iodine contrast IS NOT a contraindication to the injection of FDG.
    - 3/Risks associated with irradiation. FDG PET/CT delivers a dose of 4.4 mSv (compared with a 7.5 mSv thoraco-abdomino-pelvic scanner). The increased risk of stochastic events due to these low radiation doses is difficult to quantify. However, the risk seems very low in relation to the severity of the disease, so the risk/benefit ratio is in favor of using this imaging procedure.

- Biocollection: skin biopsy and blood tests: mild bleeding, hematoma, mild pain.

In conclusion, benefit risk ratio is favorable.

3. OBJECTIVES

3.1. Primary and secondary objectives

- **Primary objective**

In patients with R/R CTCL treated with DT-7012:

To identify the maximum tolerated dose (MTD) of intravenous dosing of single agent DT-7012, in terms of short-term safety & tolerability

- **Secondary objectives**

In patients with R/R CTCL treated with DT-7012:

1. To assess DT-7012 safety (adverse events)
2. To assess DT-7012 efficacy (response)
3. To assess pharmacokinetics of DT-7012
4. To assess DT-7012 immunogenicity
5. To assess DT-7012 pruritus (patient reported outcome)

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

6. To assess health-related quality of life (patient reported outcome)
7. To assess Time To Next Treatment
8. To assess disease control rate (DCR: ORR + stable disease)
9. To assess duration of response (DoR)
10. To assess progression-free survival (PFS)

3.2. Objective of any potential exploratory study

The objectives of the exploratory study are:

- To investigate the association of response with intensity of CCR8 expression in blood and skin at baseline and during follow-up, CCR8 expression will be studied by immunohistochemistry and multiplex immunofluorescence in lesional skin at baseline and 3 months, and by flow cytometry in peripheral blood at baseline, 1 and 3 months after inclusion
- To investigate the immune response to DT-7012 in blood and skin biomarkers (such as e.g., but not limited to, exhaustion markers PD1, TIGIT, Tregs markers CD25 and FoxP3, activation markers such as CD69, CD25 and cell markers – CD68, CD163 macrophages, CD8 T cells) will be studied by multiplex immunofluorescence in lesional skin at baseline and 3 months, and biomarkers by flow cytometry in peripheral blood at baseline, 1 and 3 months after inclusion
- To assess the molecular residual disease by high throughput sequencing of T-cell receptor genes in skin and blood (cellular and cell-free DNA) at 3 months compared to baseline

4. STUDY DESIGN

4.1. Study endpoints

4.1.1. Primary endpoint

Dose limiting toxicity (DLT) defined by any grade ≥3 toxicity (CTCAE v5.0) lasting >7 days within 3 weeks from the first DT-7012 administration, except for:

- in patients without lymphopenia at inclusion: lymphopenia considered as DLT only if ≥ grade 4
- in patients with lymphopenia at inclusion: more than 80% of reduction of lymphocytes count which is considered DLT,
- skin AE and infectious AE (very frequent in CTCL patients) considered as DLT only if ≥ grade 4 (life-threatening).

4.1.2. Secondary endpoints

- **Safety endpoints**:
    1. Adverse Events (AEs), Serious Adverse Events (SAEs), Drug related AEs, Drug related SAEs, Adverse Events of Special Interest (AESI), up to 12 months. AE will be categorized according to the CTCAE v5.0 classification.

- **Efficacy endpoints**:
    2. Objective response rate, ORR (complete response (CR) + partial response (PR)) at 3 months
       Response assessment will be performed according to the Global Response Score of the International Society for Cutaneous Lymphomas/European Organization of Research and Treatment of Cancer (ISCL/EORTC) criteria for MF/SS (Olsen, et al. 2022).
    3. PK endpoints (Maximum and trough concentration of DT-7012 at each administration, Area under the curve from time 0 to day 7 (AUC day 0-7) for the first and fourth administration, Accumulation index in terms of ratio of concentrations and AUC between the fourth and first administration),
    4. Presence of anti-drug antibodies
    5. Pruritus, assessed by a visual analogue scale (VAS) at 3, 6 and 12 months
    6. Health related Quality of life, by a standardized skin-specific questionnaire (SkinDex29) and EORTC QLQ C30 questionnaire at 3, 6 and 12 months

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

7. Time to next treatment (TTNT), defined as time from initiation of DT-7012 until the time of initiation of any systemic treatment or total-skin treatment (phototherapy or TSEB), up to 12 months. Maintenance treatment (in patients with ongoing CR or PR) and allogeneic transplantation will not be taken into account in TTNT.
8. Disease control rate (defined as complete, partial response or stable disease), at 3, 6 and 12 months
9. Duration of response (DoR) defined as the time from measurement of CR/PR (whichever is first recorded) until the date of documented recurrent or progressive disease, up to 12 months (Assessed in responder patients only)
10. Progression free survival, defined with the time from the first DT-7012 administration to progressive disease, relapse, death, or last follow-up, up to 12 months.

4.1.3. Exploratory endpoints

1. To evaluate the expression of CCR8, on tumor and reactive cells, in skin and blood, as a biomarker of response: % of positive cells within the infiltrate by IHC at baseline and 3 months, and CCR8 mean fluorescence intensity by flow cytometry in peripheral blood at baseline, 1 and 3 months after inclusion
2. To assess the immune response to DT-7012: percentages of cells positive for the following markers: exhaustion markers PD1, TIGIT, Tregs markers CD25 and FoxP3, activation markers CD69, CD25 and cell markers – CD68, CD163 macrophages, CD8 T cells) will be studied by flow cytometry in peripheral blood at baseline, 1 and 3 months after inclusion. Percentage of CCR8-positive tumor T cells and Tregs will be studied by multiplex IF in skin at baseline and 3 months
3. To assess the molecular residual disease by high throughput sequencing of T-cell receptor genes in skin and blood: % of the blood tumor clone in skin and blood (in both cellular and cell-free DNA) at 3 months compared to baseline

Definitions of responses:

- The response in skin and blood (complete/partial response/stable disease/progression) will be assessed by clinical examination + blood flow cytometry (including CD3, CD4, CD26 and CD7 markers) at baseline, and every month until 12 months thereafter, according to the ISCL/EORTC criteria.
- The response (complete/partial response/stable disease/progression) in lymph nodes and viscera will be evaluated in patients with baseline nodal (N3) or visceral involvement only, at baseline (within 3 months before baseline visit) and at 3, 6 and 12 months according to the ISCL/EORTC criteria, by CT scan or PET CT scan depending on the choice of the investigator (always the same method will be used at baseline and during follow-up in a given patient).
- ORR will be defined as partial or complete response within 3 months from the first DT-7012 administration[4].

4.2. Description of research methodology

4.2.1. Design of the study

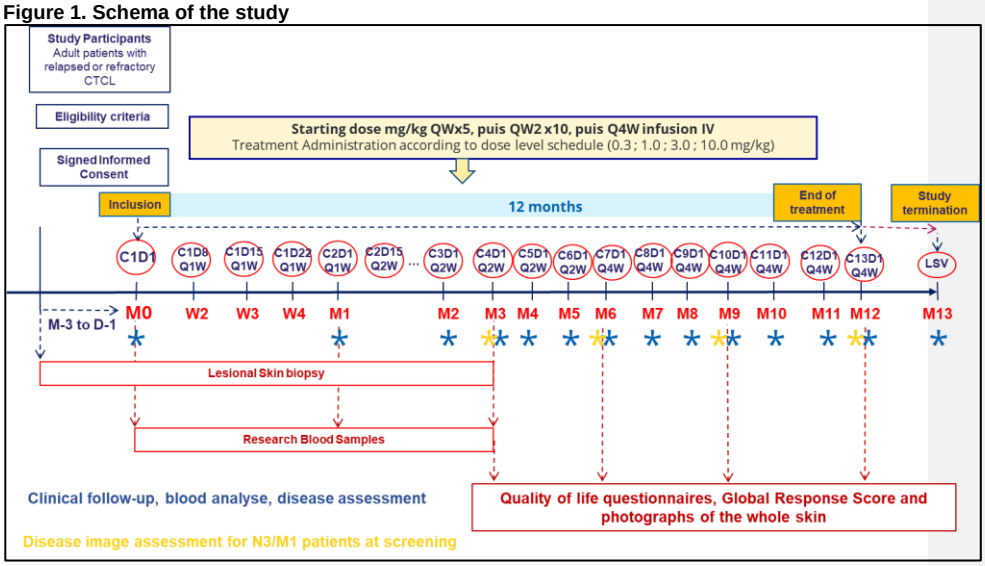
This study is a prospective, multicenter, single arm, open label phase 1 dose-escalation study, using the one-stage Bayesian Time to Event Continual Reassessment Method (TITE-CRM) design [O'Quigley 1990; Cheung 2000].
The continual reassessment method (CRM) is a model-based dose-escalation design for the phase I clinical trials, using the sequential estimation of the DLT rate at each dose level, relying on a parametric model for the dose-toxicity relationship which is sequentially updated based on accumulated DLT data during the trial, to estimated and recommend the maximum tolerated dose (MTD) [O'Quigley et al. Biometrics 1990; 46 (1): 33-48]. The CRM design has long been shown to outperform the 3+3 rule-based design in recommending the correct MTD and dosing more trial patients close to the MTD [O'Quigley J. Stat Med 1999 ;18: 2683-90; Iasonos A et al. Clin Trials 2008; 5: 465–77; Wheeler et al. BMC Med Res Method 2019; 19: 18; Chiuzan C & Dehbi HM. Clin Trials 2024; 21 (3): 350-357]. The

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

time-to-event extension of the CRM (TITE-CRM), relying on a weighted likelihood for the binomial toxicity events, allows for continuous accrual [Cheung YK & Chappell R. Biometrics 2000;56(4):1177-82]. Given the anticipated accrual rate (30 patients in 18 months, that is 1.7 patients / month on average), with regard to the primary DLT observation window (3 weeks), the TITE extension of the CRM was selected for the CITY trial design.
Specifically, the trial will follow a one-stage Bayesian TITE-CRM dose-finding design, using a one-parameter empirical dose-toxicity model (power model) to estimate the maximum tolerated dose. A linear weighting scheme will be computed from available follow-up of included patients during dose-escalation. Four candidate dose levels will be considered (0.3; 1.0; 3.0; 10.0 mg/kg) based on available data on DT-7012 and on another anti-CCR8 monoclonal antibodies (see section 2.6), initiating dose assignments at the second dose level (1.0 mg/kg). This starting dose has been chosen given existing data for available depleting monoclonal antibodies in CTCL (section 2.6).
Safety rules will be implemented, notably to ensure staggered inclusions and safe dose-escalation throughout the trial:
- No dose skipping will be allowed during dose escalation;
- For each newly escalated dose level, at least three patients must have been treated at this level prior to escalating to the next above dose level;
- For each newly escalated dose level, the first three patients treated at this dose level will be included at least one week apart;
- Dose escalation will be terminated before the planned maximum number of patients have been treated if the lower limit of the posterior 90% probability interval of the DLT rate at Dose Level 1 (DL1) exceeds the target DLT rate of 30%.

.
In case of acceptable safety profile and promising efficacy signals, a randomized study could be planned in a second step after completion of the present study.
Furthermore, intra-patient dose-escalation, after validation by the sponsor, will be allowed beyond the primary observation window, provided intra-patient absence of toxicity and available information on the dose levels above from other patients.

Figure 1. Schema of the study



4.2.2. Number of participating sites

- **Recruiting centres**

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Patients will be recruited in the dermatology departments (consultation or hospitalization) of 5 university hospital centres in France, Germany and Spain.

The relevant information and decision will be communicated to each site's team (investigators, CRA etc.) by phone followed-up by an email. This direct communication will ensure the quick and easy communication of important information, in particular regarding safety. Moreover, it is expected that almost 50% of the patients will be included in the St Louis hospital, which is a site part of the sponsor's institution.

4.2.3. Identification of participants

The participants in this Clinical Trial will be identified as follows:
Site number. (3 digits) - Sequential enrolment number for the site (4 digits) – country code
This reference number is unique and will be used for the entire duration of the study.

5. IMPLEMENTATION OF THE STUDY

The start of the clinical trial is the inclusion of the first patient (signature of the consent form).

5.1. Screening visit

The screening visit takes place between *1 day* and no later than *3 months* before the baseline visit.
Patients are informed about the study protocol at the standard care visit and the information and consent form is given at this visit. The investigator informs the patient of the study, its objectives and design (PI or another physician of department of dermatology specialty trained and declared in the study).

The physician who is responsible for the patient during the study will verify the eligibility criteria.
The screening visit will be performed at the dermatology department (consultation or hospitalization) by a specialist in dermatology from a recruiting center.

The following procedures will be performed during the screening visit by the investigator or subinvestigator (SI):

- Oral and written information about the study
- Demographics, medical history (at screening or within 3 months of baseline)
- Review of prior anti-cancer treatments and concomitant medications (at screening or within 3 months of baseline)
- Physical examination including blood pressure, pulse, weight, temperature
- Height (at screening or within 3 months of baseline)
- WHO Performance Status (at screening or within 3 months of baseline)
- Disease assessment (mSWAT, flow cytometry)
- Disease image assessment: Contrast CT-Scans or PET/CT of neck, chest, abdomen and pelvis using intravenous contrast agent. The same method should be used for repeated measurements throughout the study.

Notes:
- 1. CT is the favoured investigation for staging and evaluation of response but PET/CT is acceptable if local practice dictates this.
- 2. In case of lymph node abnormality (N1, N2 or Nx), either excisional or core biopsy of a peripheral lymph node or PET-scan in case of abnormal central nodes will be required. Central nodes are considered involved (N3) when SUV > 5 at the PET-scan.

- Non-identifying photographs of the whole skin
- Blood betaHCG pregnancy test in WOCBP
- Serologies tests
- Concomitant medications

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Whose consent must be obtained	Who informs the individuals and collects their consent	At what point the individuals are informed	At what point the consent is obtained
The individual participating in the study	The principal investigator or collaborating physician declared and trained in the study (dermatologist)	Screening visit (standard of care visit)	Within 3 months before or during baseline visit, after a period of reflection of at least 24 hours.

5.2. Baseline visit/Inclusion visit

Following informed written consent, the baseline visit will be performed at clinical investigation center.
Ideally, the baseline visit should be C1D1 of treatment.

At baseline or within 1 month prior to baseline:

- Signature of informed consent
- Blood burden: absolute counts of CD3, CD4, CD8, CD4+CD7- and CD4+CD26-
- Disease image assessment: Contrast CT-Scans or PET/CT of neck, chest, abdomen and pelvis using intravenous contrast agent. The same method should be used for repeated measurements throughout the study.

Notes:
- 1. CT is the favoured investigation for staging and evaluation of response but PET/CT is acceptable if local practice dictates this.
- 2. In case of lymph node abnormality (N1, N2 or Nx), either excisional or core biopsy of a peripheral lymph node or PET-scan in case of abnormal central nodes will be required. Central nodes are considered involved (N3) when SUV > 5 at the PET-scan.

- Non-identifying photographs of the whole skin
- Skin assessment based on mSWAT

The following procedures will be performed during the baseline visit:

- Check inclusion and exclusion criteria
- Review of concomitant medications, date of start and end, related medical reason
- Physical examination including blood pressure, pulse, weight, temperature
- WHO Performance Status
- Skin assessment based on mSWAT *if not done in the past month*
- Blood tests: CBC with differential, Na/K/Ca/P/glycemia/albumin/creatinine, AST, ALT, gGT, alkaline phosphatases, bilirubin (total, free and conjugated),
- Blood burden: absolute counts of CD3, CD4, CD8, CD4+CD7- and CD4+CD26 *if not done in the past month*
- Disease image assessment, *if not done in the past month*
- Photographs of the skin *if not performed in the past month*
- Adverse events reporting
- Pregnancy test for WOCBP (blood or urine)
- Treatment administration.

To be performed at baseline only for research purposes after signed informed consent:

- Formalin fixed-paraffin embedded (FFPE)
- Frozen 4-mm punch biopsy from lesional skin (optional, in participating sites only)
- Tape stripping: noninvasive skin sampling (optional, in participating sites only)
- Blood samples for research (see section 5.11)
- Patient-reported health related quality of life, assessed by the EORTC-QLQ-C30 and Skindex-29 questionnaires + VAS pruritus

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

5.3. Treatment visits

The IMP will be administered first at baseline (=C1D1), then Q1W for a total of 5 injections, then Q2W for 10 injections, then Q4W until progression or unacceptable AE. The injections will be performed at the Centres d'Investigation Clinique in France or in a structure approved for first-in human administration and the patient may be hospitalized depending on the local organisation.

Following DT-7012 infusions, participants must be observed for 6 hours post infusion for potential infusion-related reactions for the first injection, then they will be observed for 2 hours after each subsequent infusion.
For the first 3 infusions, a phone call at 24h will be made to check on the patient.

Pre- and post-administration vitals monitoring: Heart rate, blood pressure (systolic and diastolic), body temperature and respiratory rate should be measured at each visit before administration (within 30 min of administration) and after administration during the observation period.

Besides the treatment administration, blood samples will be taken at these visits for PK and ADA analysis (see section 5.11).

5.4. Follow-up ± treatment visits

The follow-up visits may or may not coincide with the treatment visits. Ideally, Month 3 and Month 6-9-12 evaluations should be combined with treatment cycles during the same day hospital stay.

See section 5.3 for schedule of treatment administration and blood samples for PK and ADA analysis.

Every month from baseline:

- Continuous recording of any adverse event, date of start and end, CTCAE v5.0 grade, considered related to study treatment or not
- Review of concomitant medications, date of start and end, related medical reason
- Physical examination including blood pressure, pulse, weight, temperature
- WHO Performance Status
- Blood tests: CBC with differential, Na/K/Ca/P/glycemia/albumin/creatinine, AST, ALT, gGT, alkaline phosphatases, bilirubin (total, free and conjugated),
- Pregnancy test (urine or blood) in WOCBP before each infusion
- Disease assessment:
  - Skin assessment based on mSWAT
  - Blood burden (flow cytometry): absolute counts of CD3, CD4, CD8, CD4+CD7- and CD4+CD26-.

Additionally, some procedures are performed at specific visits.

At week 1 only:
- Blood collection for research (see section 5.11)

At M1 and M3 only:
- Blood collection for research (see section 5.11)
- Tape stripping: noninvasive skin sampling (optional, in participating sites only) (see section 5.11)

At M3 only:
- Formalin fixed-paraffin embedded (FFPE) (see section 5.11)
- Frozen 4-mm punch biopsy from lesional skin (optional, in participating sites only) (see section 5.11)

At M3, M6, M9 and M12 only:
- Disease image assessment must be **performed at 3, 6, 9 and 12 months in patients with baseline nodal (N3) or visceral involvement, or in case of clinically suspected nodal or visceral progression.** Contrast CT-Scans or PET/CT of neck, chest, abdomen and pelvis using

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

intravenous contrast. The same method should be used for repeated measurements throughout the study. Notes: 1. CT is the favored investigation for staging and evaluation of response but PET/CT is acceptable if local practice dictates this. 2. In case of lymph node abnormality (N1, N2 or Nx), either excisional or core biopsy of a peripheral lymph node or PET-scan in case of abnormal central nodes will be required. Central nodes are considered involved (N3) when SUV > 5 at the PET-scan.
- Patient-reported health related quality of life, assessed by the EORTC-QLQ-C30 and Skindex-29 questionnaires, pruritus VAS.
- Global response score
- Whole skin photographs

5.5. Visit for Suspected progressive disease

- Physical examination (including weight, blood pressure, pulse, temperature)
- WHO Performance Status
- Adverse event assessment according to CTCAE version 5.0
- Concomitant medications
- Quality of life questionnaires
- Haematology and serum chemistry
- Disease assessment: Skin assessment based on mSWAT; Blood burden (flow cytometry): absolute counts of CD3, CD4, CD8, CD4+CD7- and CD4+CD26-.
- Disease imaging
- Global response score

5.6. Last study visit

The following assessments will be performed within 30 days after last protocol treatment administration (after 12 months from treatment start or earlier in case of progressive disease or the occurrence of a withdrawal criterion).

- Physical examination (including weight, blood pressure, pulse, temperature)
- WHO Performance Status
- Adverse event assessment according to CTCAE version 5.0
  - If at the time of this visit, patient has ongoing severe toxicities (grade 3 and 4), follow up of adverse events related to DT-7012 should be provided for these toxicities until recovery to grade 1 or baseline.
- Concomitant medications
- Quality of life questionnaires + pruritus VAS
- Haematology and serum chemistry
- If disease progression has been properly documented, disease evaluation must not be repeated.
- Disease image assessment **in patients with baseline nodal (N3) or visceral involvement, or in case of clinically suspected nodal or visceral progression**
- Blood collection for research (see section 5.11)

5.7. Early termination visit

The following assessments will be performed within 30 days after termination.

- Physical examination (including weight, blood pressure, pulse, temperature)
- WHO Performance Status
- Adverse event assessment according to CTCAE version 5.0
  - If at the time of this visit, patient has ongoing severe toxicities (grade 3 and 4), follow up of adverse events related to DT-7012 should be provided for these toxicities until recovery to grade 1 or baseline.
- Concomitant medications
- Haematology and serum chemistry

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

5.8. Expected length of participation and description of the chronology and duration of the study

Duration of enrolment period	18 months
Maximum period between screening and baseline	3 months
The length of participation for participants, of which:	
• Treatment duration	12 months
• Duration of follow-up period	1 month
Total study duration	31 months

5.9. Table or diagram summarizing the chronology of the study

	Screening	Baseline	Monthly follow-up ±treatment visits	Study end M13	Suspected progressive disease
	(Within 3 months to 24 hours prior to baseline)	(Cycle 1 Day 1)	(± 3 days)	(± 3 days)	(± 3 days)
Information	R				
Verification of inclusion and exclusion criteria	R				
Signing consent		R			
Height	X (a)				
Demographics, medical history	X (a)				
Prior anti-cancer treatments	X (a)				
Serologies	x				
Treatment administration DT-7012		R	see section 5.3 for schedule of administration		
Concomitant medications	R	R	R	R	R
Physical examination (blood pressure, pulse, weight, temperature)	X	x	x	x	x
Adverse events according to CTCAE v 5.0 (continuous record)		R	R	R	R
WHO performance status	X (a)	x	x	x	x
Pregnancy test (WOCBP)	X**	x	x	x	
Haematology		x	x	x	x
Serum chemistry		x	x	x	x
Disease assessment (mSWAT, standard CTCL flow cytometry)	X	X if not performed <1 month	x	x	x
Disease imaging assessment in patients: CT-scan or PET/CT	X	X if not performed <1 month	R* (M3, M6, M9, M12 only)	X*	x

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Global Response Score			x (M3, M6, M9, M12 only)	x	x
Clinical whole skin photographs	x	X if not performed <1 month	x (M3, M6, M9, M12 only)		
Skindex-29 and EORTC QLQ C30 QOL questionnaires, VAS pruritus		R	R (M3, M6, M9, M12 only)	R	
Lesional skin biopsies (1 FFPE and 1 optional frozen skin biopsy)		R	R (M3 only)		
Blood for research (see section 5.11)		R	R (week 1, M1, M3 only)		
PK and ADA analysis		R	R (See section 5.11)	R	
Tape stripping		R	R (M1, M3 only)		

Inclusion and C1D1 should ideally be combined during the same day hospital stay.
Month 3 and Month 6-9-12 evaluations should ideally be combined with treatment cycles during the same day hospital stay.
**Optional: only if N3/M1 at baseline. (a): results obtained within 3 months of baseline are accepted.*
X** Blood betaHCG

5.10. Distinction between standard care and study

TABLE: "Standard care" vs. "additional interventions" required specifically for the study

Interventions, procedures and treatments carried out for research purposes	**Interventions, procedures and treatments associated with <u>standard care</u>**	**Interventions, procedures and treatments added for <u>research purposes</u>**
Treatments		DT-7012 IV infusions
Visits		Baseline (=C1D1) then Q1W for X5 then Q2W for X10 then Q4W for max total 12 months and EOT: 21 visits
Blood samples	Standard flow cytometry, hematology and blood chemistry	-Blood samples for immune monitoring at baseline, week 1, M1, and 3 months (PBMC and plasma + Streck tube and PAXgene tube) -PK and ADA samples -Monthly B-HCG
Imaging	Contrast CT-scan or PET/CT at <u>screening,</u> baseline, progression ~~and end of treatment~~	Contrast CT-scan or PET/CT at 3-6-9-12 months ~~and end of treatment~~ only **in patients with N3/M1 stage at baseline**
Skin biopsies		- 2 skin biopsies 4-mm at baseline and at 3 months (1 mandatory FFPE and 1 optional frozen)

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Skin tape stripping		- Tape-stripping at baseline, M1 and M3 (optional)
Questionnaires		Skindex-29, EORTC-QLQ-C30 and VAS pruritus at baseline, M3, 6, 9, 12 months and study end
Clinical photographs	Within 1 month prior to baseline, and M3, 6, 9, 12 months	

5.11. Biological samples collection – for research

For each patient, the following samples will be taken.

For more details on processing and storage, please see laboratory manual.

Blood samples for PK analysis:
Number of tubes per timepoints: 1x 3.5ml silica-coated tubes for serum

Timepoints:

- Infusions 1 to 5: prior to IMP infusion (pre-dose), at the end of the IMP infusion, and at 30 min (± 5 min), 1 h (± 10 min), 4 h (± 30 min), and 6 h (± 30 min) after the end of infusion (post-infusion). On D2, a PK sample will be collected at 24 h (± 3 h) post-infusion only on cycle 1.

- Infusions 7, 9, 11, 13, 15, 16, 17, 18, 19, 20, 21: prior to IMP infusion (pre-dose) and at 1 h (± 10 min) after the end of infusion (post-infusion).

- Last visit (M13): any time during the visit (collection time must be recorded).

Blood samples for ADA analysis:
Number of tubes per timepoints: 1x 5ml silica-coated tubes for serum

Timepoints:

- Infusions 1 to 5: prior to IMP infusion (pre-dose).

- Infusions 7, 9, 11, 13, 15, 16, 17, 18, 19, 20, 21: prior to IMP infusion (pre-dose)

- Last visit (M13): any time during the visit (collection time must be recorded).

Blood samples for soluble biomarkers:
Number of tubes per timepoint: 1x 5ml silica-coated tubes for serum

Timepoints: baseline, week 1, M1, M3

Blood samples going to Inserm U1160:

Number of tubes per timepoint: 7x 7ml EDTA tubes, 1x 2,5 ml Paxgene tube, 1x 10 ml Streck tube

Timepoints: baseline, M1, M3

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

For skin samples: 1 skin biopsies 4-mm at baseline and at 3 months (1 mandatory FFPE), skin trap stripping

Skin biopsy
FFPE skin block will be stored on site.

- FFPE blocks will then be sent to INSERM U1312 Bordeaux (France) under the supervision of Professor Anne Pham-Ledard for multiplex IF analysis and stored at RT.
- FFPE scrolls will be sent to the department of pathology at INSERM U976 Saint-Louis hospital under the supervision of Professor Battistella for clonality analysis by high throughput sequencing of T-cell receptor genes.
- Skin (slides) will be sent to INSERM U955 Henri-Mondor (France) and analysed and stored under the supervision of Professor Nicolas Ortonne for CCR8 IHC Study of prognostic factors, discovery of therapeutic targets, pathophysiological mechanisms of response or resistance to treatment.

Frozen samples in participating sites only.

Skin tape stripping (Performed in participating site only)

Number of samples per timepoint: 8 tape strips from Lesional areas and 8 from Non Lesional areas.

Timepoints: predose (baseline), M1 and M3.

Samples taken as part of the study will be stored in a biological sample collection. The samples will be pseudonymised.

At the end of the study (including the analysis of the samples), the samples will be kept for a maximum of 30 years.

The samples may be used for further analysis not described in the initial protocol but which may be useful for investigation of CTCL in light of advances in scientific knowledge, provided the participant is informed and does not oppose this, as stated in the information note/consent form.

The samples will be pseudonymised with a direct connection with the anonymisation process of recruited patients. The sample will be directly connected to the patient using initials and date of birth.

The sample collection Lymphoteq is already declared to the ministry of research [and to the director of the competent regional healthcare authority if the entity is a health establishment] (Article L. 1243-3 of the *Code de la Santé Publique* [French Public Health Code]).

Type of sample	**Quantity**	**Storage location (name and entity)**	**Supervisor of the sample collection (name and entity)**	**Purpose of the sample collection**	**Storage duration**	**End use/Future (e.g., destruction) after the end of the CT**
Blood for PK and ADA analysis	In total: 54 tubes x 3.5 ml for PK (see above for details) In total: 17 tubes x 5 ml for	On recruiting site then sent to 360° Biolabs	Hélène Lelièvre (Domain therapeutics)	PK and ADA analysis	Study period	Sent back to Sponsor after analysis. Kept for max 30 years.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

	ADA (see above for details)					
Blood for plasma	1 tube x 5ml At baseline, week 1, M1 and M3	On recruiting site then sent to Domain therapeutics	Hélène Lelièvre (Domain therapeutics)	Soluble biomarkers	Study period	Sent back to Sponsor after analysis. Kept for max 30 years.
Blood (PBMC, plasma, and whole blood)	7x 7ml EDTA tubes, 1x 2,5ml PAXgene tube, 1x 10ml Streck tube At baseline, 1 month and 3 months	On recruiting site and then sent to Inserm U1160, St Louis Hospital	Prof Hélène Moins	Study of prognostic factors, discovery of therapeutic targets, pathophysiological mechanisms of response or resistance to treatment	Study period then 30 years	Kept for max 30 years
Skin (block)	1 skin biopsy block (4-mm) At baseline and at 3 months (FFPE)	Local storage, then sent to Prof Pham-Ledard (INSERM U1312 Bordeaux).	Prof Anne Pham-Ledard (INSERM U1312 Bordeaux (France))	Multiplex IF Study of prognostic factors, discovery of therapeutic targets, pathophysiological mechanisms of response or resistance to treatment	Study period then 30 years	Kept for max 30 years
Skin (scrolls)	Scrolls from 1 skin biopsy block (4-mm) At baseline and at 3 months (FFPE)	Local storage, then sent to Prof Battistella (INSERM 976 St Louis, Paris).	Prof Battistella (INSERM U976 Saint-Louis hospital)	Clonality analysis by high throughput sequencing of T-cell receptor genes	Study period then 30 years	Kept for max 30 years

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Skin (slides)	Slides from 1 skin biopsy (4-mm) At baseline and at 3 months (FFPE)	Local storage then sent to Prof Ortonne (INSERM U955 Mondor)	Prof Ortonne (INSERM U955 Mondor)	CCR8 IHC Study of prognostic factors, discovery of therapeutic targets, pathophysiological mechanisms of response or resistance to treatment	10 years	Kept for max 10 years
Skin (frozen)	1 optional skin biopsy block (4-mm) At baseline and at 3 months (frozen)	Local storage	In each center under the responsibility of the investigator	Study of prognostic factors, discovery of therapeutic targets, pathophysiological mechanisms of response or resistance to treatment	30 years	Kept for max 30 years
Skin (tape-strips)	8 tape strips from Lesional areas and 8 from Non Lesional areas. Collected at predose (baseline), M1 and M3.	Local storage then sent to Domain therapeutics	In each center under the responsibility of the investigator, then Hélène Lelièvre (Domain therapeutics)	Study of prognostic factors, pathophysiological mechanisms of response or disease progression	30 years	Kept for max 30 years

6. ELIGIBILITY CRITERIA

6.1. Inclusion criteria

To be eligible for study entry patients must meet all of the following criteria
1. Adult patients (≥18 years) with no upper age limit
2. Confirmed diagnosis of mycosis fungoides or Sezary syndrome
3. Stage IB to IVB in the ISCL / EORTC classification
4. Relapsed or refractory (no response) after at least two systemic treatments
5. ECOG performance status 0-1
6. Adequate liver function:
   - Total bilirubin ≤ 1.5 x ULN, or Direct bilirubin ≤ 1.5xULN if total bilirubin is >1.5xULN, or total bilirubin >1.5 xULN but elevated total bilirubin is attributed to Gilbert's syndrome or to histologically-proven liver involvement by CTCL.
   - Aspartate aminotransferase (AST) and alanine aminotransferase (ALT) ≤ 2,5 x ULN, unless elevated to up to 5 x ULN due to CTCL
7. Adequate hematological function:

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

- Absolute neutrophil count of ≥ 1.5 G/L without G-CSF support for at least 7 days
- Platelet count of ≥ 75 G/L without platelet transfusion within 7 days
- Hemoglobin ≥ 9 g/dL without RBC transfusion within 7 days

8. Adequate renal function: creatinine clearance calculated by Cockcroft & Gault formula of ≥ 50 mL/min
9. HBV: negative blood HBs Ag or blood HBV DNA. Vaccinated patients may be included. Patients with HBc antibody may be included if HBV DNA is negative
10. HCV: negative HCV serology, or negative HCV RNA if HCV serology is positive
11. HIV: negative HIV serology
12. Negative serum or urinary pregnancy test within 7 days or at baseline prior to study treatment in women of childbearing potential
13. Patients must agree to use a highly effective contraceptive method from inclusion until:
    - If the patient is a male: at least 6 months after the last dose of DT-7012. Men must refrain from donating sperm during this same period
    - If patient is a female of childbearing potential: at least 6 months after the last dose of DT-7012
16. Patients must have the following minimum wash-out from previous treatments:
    - 12 weeks for total skin electron beam irradiation,
    - 4 weeks for monoclonal antibodies
    - 3 weeks for local radiation therapy, systemic cytotoxic anticancer therapy, treatment with other anti-neoplastic investigational agents
    - 3 weeks for systemic retinoids, interferons, vorinostat, romidepsin, fusion proteins
    - 3 weeks for phototherapy
    - 2 weeks for topical therapy (including steroids, retinoids, nitrogen mustard or imiquimod). Topical steroids and oral steroids (10 mg prednisone equivalent/day maximum) are allowed, if the patient has been on a stable dose with stable symptoms for at least 4 weeks prior to study entry.
15. Patient covered by any social security system (registered or being a beneficiary of such a scheme)
16. Signed informed consent

6.2. Exclusion criteria

Patients meeting any of the following criteria will not be included in the study:

1. Known central nervous system involvement by CTCL
2. Participation in any study of a health product within 30 days prior to study entry
3. Patients with a history of other malignancies during the past three years (*except for: non-melanoma skin cancer, lymphomatoid papulosis, curatively treated localized prostate cancer, curatively treated localized breast cancer, resected thyroid cancer, biopsy proven cervical intraepithelial neoplasia or cervical carcinoma in situ, which are not considered exclusion criteria).*
4. Uncontrolled congestive heart failure (New York Heart Association Classification 3 or 4), or angina, myocardial infarction, cerebrovascular accident, transient ischemic attack within 1 month prior to study entry
5. Any severe acute or chronic medical or psychiatric condition
6. Patients with immunodeficiency
7. Any active systemic viral, bacterial, or fungal infection requiring systemic antibiotics within 1 week prior to first study drug dose
8. Chronic use of systemic corticosteroids of prednisone or equivalent >10 mg per day of prednisone equivalent, for a chronic condition (washout of 8 days from start of treatment is accepted)
9. Other immunosuppressive therapies are also excluded, (washout of 7 days from start of treatment is accepted)
10. Autologous HSCT within 100 days prior to DT-7012 infusion
11. Prior allogeneic HSCT
12. Prior solid organ transplantation
13. Patient with history of confirmed progressive multifocal leukoencephalopathy
14. Known or suspected allergies, hypersensitivity, or intolerance to DT-7012 or its excipients
15. Pregnant or breast-feeding woman, or desire (for both man and woman participant) to conceive a child within 6 months after end of treatment
16. Patient under guardianship or curatorship and protected adults or unable to consent
17. Coagulation disorder contra indicating intravenous infusion
18. History of anaphylactic reaction following vaccination or immunotherapy

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

6.3. Contraception during the study

Contraception during the study according to the recommendations related to contraception and pregnancy testing in clinical trials – Clinical Trial Coordination group CTCG version 1.2 (07/03/2024).
A highly effective birth control method must be used from inclusion until 6 months after the last dose of treatment for female participant of childbearing potential and male participant with a female partner of childbearing potential.

6.3.1. Definition of women of childbearing potential and of fertile men

Woman is considered of childbearing potential (WOCBP), i.e., fertile, following menarche and until becoming post-menopausal unless permanently sterile. Permanent sterilisation methods include hysterectomy, bilateral salpingectomy and bilateral oophorectomy.
A postmenopausal state is defined as no menses for 12 months without an alternative medical cause. A high follicle stimulating hormone (FSH) level in the postmenopausal range may be used to confirm a post- menopausal state in women not using hormonal contraception or hormonal replacement therapy. However, in the absence of 12 months of amenorrhea, a single FSH measurement is insufficient.

Man is considered fertile after puberty unless permanently sterile by bilateral orchidectomy.

6.3.2. Pregnancy control requirements

Pregnant and breast-feeding women cannot participate in the study.
All WOCBPs must have a highly sensitive negative serum pregnancy test prior to treatment and must agree to maintain highly effective contraception or using an effective method of birth control during treatment.

In every case of delayed menses (more than one month between periods), it is strongly recommended to confirm the absence of pregnancy. This recommendation also applies to WOCBPs with infrequent or irregular menstrual cycles.

Monthly pregnancy testing (urine test) will be performed in WOPCBP, from inclusion to 2 months after end of treatment.

6.3.3. Birth control methods which may be considered as highly effective

For all WOCBP, methods that can achieve a failure rate of less than 1% per year when used consistently and correctly are considered as highly effective birth control methods. Such methods include:

- Combined (estrogen and progestogen containing) hormonal contraception associated with inhibition of ovulation (Oral, Intravaginal, Transdermal);
- Progestogen-only hormonal contraception associated with inhibition of ovulation (Oral, Injectable, Implantable);
- Intrauterine device (IUD);
- Intrauterine hormone-releasing system (IUS);
- Bilateral tubal occlusion;
- Vasectomised partner.

For men:

- Condom

6.4. Recruitment procedure

Patients will be recruited in the Dermatology departments of the 5 recruiting centers (Bordeaux, Paris, Lyon, Berlin, Madrid).
In the CUTALLO clinical trial (6) at the 2 French centers, 67 patients with advanced stage CTCL (stages IIB to IV) were enrolled over 5,5 years (13.4 patients per year).
Therefore, we are confident that we can enroll 30 patients over a 18-month period.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

	Number of participants
Total number of participants to be included	30
Number of centres	5
Enrolment period (months)	18
Number of participants/centre and per month	0,3

6.5. Termination rules

6.5.1. Termination rules for individual participants

6.5.1.1. Criteria and procedures for temporary suspension of the study treatment and continuation of the participation in the study

Temporary suspension of the treatment is allowed at all stages of the treatment, especially to recover from adverse events.
The treatment can be resumed once the adverse event has been resolved.
Maximum suspension acceptable to resume and continue the trial schedule is 14 days.

If, during the course of his/her participation in the study, the participant presents one of the following adverse events, the study product must be temporary discontinued but the participant will continue to be monitored for the study:

- Suspected progressive disease
- Suspected CTCAE grade ≥ 3 treatment related SAE except for:
  - in patients without lymphopenia at inclusion: if lymphopenia ≥ grade 4, treatment should be discontinued
  - in patients with lymphopenia at inclusion: if more than 80% of reduction of lymphocytes count which is considered DLT, treatment should be discontinued
  - Skin AE and infectious AE: treatment should be discontinued if ≥ grade 4 (CTCAE).

In all cases, the investigator must document the reason for suspending and resuming the treatment in the participant's source file and the case report form (CRF).

The investigator must:
- Document the reason(s)
- Collect the assessment criteria at the time of ending participation in temporary suspension of the study, if the participant agrees treatment.
- Schedule a dedicated follow-up for the participant, particularly in case of a serious adverse event.

As a general point of view, dose adaptation or the non-administration of treatment due to the occurrence of toxicity does not lead to premature discontinuation of the study.

In the case of serious adverse events, the investigator must notify the sponsor and follow up the participant until the severe adverse events are resolved. Notification of a serious adverse event must be sent by email (eig-vigilance.drc@aphp.fr) to the sponsor.

The Data Safety Monitoring Board can specify and/or validate the follow-up methods.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

6.5.1.2 Criteria and procedures for definitive discontinuation of the study treatment and continuation of the participation in the study

If, during the course of his/her participation in the study, the participant presents one of the following adverse events or toxicity of the study product or auxiliary medicinal products that would be in the investigator's opinion, detrimental to the patient's well-being, such as the following events (non-exhaustive list), then the study product must be definitively discontinued but the participant will continue to be monitored for the study (except in the situations described in section 6.5.1.3).

The following conditions require patient discontinuation from study treatment:

- Clinically relevant disease progression according to EORTC-ISCL-USCLC criteria

Note: as per mSWAT, B-class or lymph nodes may fluctuate with time. In order to ensure that a patient is not prematurely withdrawn from the study and after consultation with medical monitor, investigators are allowed to maintain patients under treatment until next evaluation, i.e., 1 month later for blood and/or skin PD and 3 months later in case PD in lymph nodes and/or viscera. If PD is confirmed, then patients should be withdrawn from protocol treatment.

- Dose limiting toxicity
- AE or intercurrent illness that, in the opinion of the investigator, precludes further therapy
- Investigator's decision to stop the treatment to the patient e.g., symptomatic clinical status deterioration without objective evidence of disease progression or in presence of evidence of unequivocal disease progression
- Non-compliance with study treatment or procedure requirements
- Sexually active patients who refuse to use medically accepted adequate birth control methods as described in the patient selection criteria during the course of the study
- A female patient inadvertently becomes pregnant
- Request by regulatory agencies for termination of treatment of an individual patient or all patients under the protocol.
- Occurrence of a new malignancy except in situ carcinoma, basal cell carcinoma or cutaneous spindle cell carcinoma

In case of serious adverse events, the investigator must notify the sponsor and follow up the participant until the serious adverse events are resolved.

If a participant withdraws prematurely the study treatment, and if the participant agrees, the procedure and schedule will be performed according to the protocol.

6.5.2 Criteria and procedure for study discontinuation

- Participants may withdraw their consent to participate and exit the study at any time and for any reason.
- Participant lost to follow-up: the participant cannot be located. The investigator must make every effort to reconnect with the participant (and document his attempts in the source file), at least to determine whether the participant is alive or dead

If a participant exits the study prematurely (consent withdrawal, lost to follow-up), any data collected prior to the date of premature exit may still be used.

- If a participant exits the study prematurely, and if the participant agrees, the collection of the data required by the protocol (primary endpoint, secondary endpoints, safety assessment) will be performed as per M1 (NB: this must be stated in the information and consent form). The patient will receive best standard of care according to his/her disease stage and previous medical history.
- A premature exit from the trial will not affect the participant ongoing care and access to standard of care
- In case of serious adverse events, see the appropriate section on SAE reporting.

The case report form must list the various reasons why the participant has discontinued the study:

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

☐ Consent withdrawal of the participant
☐ Lost to follow-up

6.5.3 Procedures for replacing participants if applicable

All patients included and having received at least one dose of treatment will be analysed.
There will be no replacement in case of premature study drop out and not replaced.

6.5.4 Termination rules for the clinical trial: full or partial discontinuation of the clinical trial

AP-HP as sponsor or the Competent Authority can prematurely discontinue all or part of the study, temporarily or permanently, further to the recommendations of the Data Safety Monitoring Board in the following situations:

- first of all, if suspected unexpected serious adverse reactions (SUSARs) are observed requiring a reassessment of the benefit-risk ratio for the study

In case of early and temporary discontinuation of the study with inclusions suspended, or in case of early definitive discontinuation of the entire study, the DSMB will meet within 72h and decide whether the patients currently receiving the study treatment should immediately discontinue the study treatment or be allowed to complete their treatment.

Similarly, AP-HP, as the sponsor, or the Competent Authority may decide to prematurely discontinue the study due to unforeseen issues or new information about the product, in light of which the objectives of the study or clinical program are unlikely to be achieved.

AP-HP as sponsor reserves the right to permanently suspend enrolment at any time if it appears that the inclusion objectives are not met.

In all cases in which a study is discontinued, the care options are specified for participants currently enrolled in the study as described in the chapter 6.5 Termination rules. The participants included in the study must be monitored until the end of their participation, as set forth in the protocol.

If the study is prematurely discontinued for safety reasons, the decision and justification will be provided by the sponsor (AP-HP) to the Competent Authority and to the Ethics Committee without undue delay but not later than in 15 days of the date of the temporary halt or early termination. It shall include the reasons for such action and specify follow-up measures, along with recommendations from the Data Safety Monitoring Board in the case of substantial modification.

7. TREATMENT ADMINISTERED TO STUDY PARTICIPANTS

7.1. Description of the investigational medicinal product(s)

7.1.1. Investigational medicinal product

DT-7012 drug product is provided as a sterile concentrate for solution for infusion containing 100 mg of DT-7012 in a 10 mL amber glass vial. Each vial contains an extractable volume of 4 mL of the 25 mg/mL of DT-7012 liquid formulation in 20 mM histidine/histidine hydrochloride (His/His-HCl) solution, 8% (w/v) sucrose, 0.02% (w/v) P188, with a pH of 5.5.
Boxes containing 1 or 4 vials will be made available for this trial.

DT-7012 drug product must be stored at -20°C ± 5°C protected from light, in an alarmed and monitored freezer, with a back-up equipment, in a secure area with restricted access to authorized trial staff.

The solution for infusion containing DT-7012 drug product must be prepared under aseptic conditions in compliance with requirements for parenteral preparation. Aseptic conditions mean the standard

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

hospital conditions for injectable preparations to prevent any contamination by microorganisms, and thus ensure that the sterility of DT-7012 solution for infusion is maintained.
DT-7012 must be diluted in glucose 5% prior to i.v. administration to trial participants.

Once thawed, DT-7012 drug product must not be re-frozen.

Further information on DT-7012 drug product preparation is given in the trial-specific Pharmacy Manual.

Dose level	Dose-schedule
DL1	0.3 mg/kg Q1W for 5 infusions then Q2W for 10 infusions then Q4W until end of treatment
DL2	1 mg/kg Q1W for 5 infusions then Q2W for 10 infusions then Q4W until end of treatment
DL3	3 mg/kg Q1W for 5 infusions then Q2W for 10 infusions then Q4W until end of treatment
DL4	10 mg/kg Q1W for 5 infusions then Q2W for 10 infusions then Q4W until end of treatment

Patients will be assigned to a dose level following the TITE-CRM dose escalation design, with 1 mg/kgas starting dose (see section 12 for details), based on the primary toxicity observation window of 3 weeks after the first administration. Beyond the 3-week primary window, intra-patient dose-escalation may be performed conditional on the tolerability of the treatment intra-patient and on the safety data available on higher dose levels, inter-patient; in any case, intra-patient dose-escalation will be assessed and proposed by the study steering committee, and will be validated by the sponsor.



- No dose skipping will be allowed during dose escalation.
- For each newly escalated dose level, at least three patients must have been treated at this level prior to escalating to the next above dose level.
- For each newly escalated dose level, the first three patients treated at this dose level will be included at least one week apart.

Following DT-7012 infusions, participants must be observed for 6 hours post infusion for potential infusion-related reactions for the first infusion, then they will be observed for 2 hours for the subsequent infusions. For the first 3 infusions, a phone call at 24h will be made to check on the patient.

Pre- and post-administration vitals monitoring: Heart rate, blood pressure (systolic and diastolic), body temperature and respiratory rate should be measured at each visit before administration (within 30 min of administration) and after administration during the observation period.

The DSMB may recommend an IRR prophylaxis regimen after review of the safety data generated during the dose escalation with DT-7012.

7.2. Description of traceability elements accompanying the investigational medicinal product(s)

All prescriptions and administrations of DT-7012, with doses and dates/hours, will be recorded in the local medical file of the patient as it is done in the standard of care context.
The preparation corresponding to each administration will be traced in the center's pharmacy documentation.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

7.3. Description of Auxiliary medicinal product(s) (treatments required to conduct the study)

According to the Regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014, an auxiliary medicinal product is a medicinal product used in the context of a clinical trial but not as investigational medicinal product.

Auxiliary medicinal product used in this research are:

Methylprednisolone
If a patient experiences a Grade ≥3 IRR, the patient will receive methylprednisolone at the next infusion as a prophylaxis in a 30 minutes infusion 1 hour before DT-7012 infusion.
All French, German and Spanish commercial specialties may be used. Instructions for use must comply with the corresponding SPC.

Anti-pyretic and anti-histamine may be used in case of adverse event.
Oral or IV Paracetamol 1000 mg
Oral Cetirizine 10 mg

^{18}FDG tracer
All the specialties commercially available in France, Spain and Germany are likely to be used. Instructions for use must comply with the SPC of the specialty used.

The auxiliary medicinal products will be supplied by each of the centers (locally available drugs).

7.4. Authorised and prohibited treatments (medicinal, additional medicinal, surgical), including rescue medications

All treatments that the investigator considers necessary for a patient's welfare may be administered at the discretion of the investigator in keeping with the community standards of medical care.

All concomitant medication will be recorded on the electronic case report form (eCRF) including all prescriptions, over-the-counter, herbal supplements, and IV medications and fluids. If changes occur during the study period, documentation of drug dosage, frequency, route, and date will also be included in the eCRF.

7.4.1. Prohibited medications

- Other investigational drugs,
- Systemic immunosuppressive medications other than stable dosage of oral steroids (10 mg prednisone equivalent/day maximum), if the patient has been on a stable dose with stable symptoms for at least 4 weeks prior to study entry, to control the cutaneous symptoms of CTCL.
- Anti-neoplastic therapy,
- Vaccination containing living viruses

7.4.2. Permitted medications

- In the absence of compatibility studies, DT-7012 should not be infused concomitantly in the same intravenous line with other medicinal products.

- Pre-medication with a non-NSAID anti-pyretic, methylprednisolone and anti-histamine may be used in case of adverse event.

- Topical (maximum level: high potency, excluding clobetasol), intra-articular, intraocular, inhalation or nasal corticosteroids or low doses of systemic corticosteroids (equivalent to ≤ 10 mg prednisone per day) may be used during treatment with DT-7012; however, the risk of serious infection and/or viral reactivation may be higher in case of concomitant administration with systemic immunosuppressive agents. Patients should be monitored for signs and symptoms of infection and treated promptly.

7.5. Methods for monitoring compliance with the treatment

There will be no compliance problem because the administrations will be exclusively intra-hospital.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

7.6. Description of the treatment modalities

The practical procedures for product preparation by hospital pharmacies and administration are described in the pharmacy manual.

8. EFFICACY ASSESSMENT

8.1. Evaluation of efficacy

Disease evaluation relies on previous recommendations (ISCL / EORTC criteria 2021)[15]

8.1.1. Response in Skin

ISCL/USCLC/EORTC response in skin for MF/SS

Response	Definition	Additional criteria
CR*	100% clearance of skin lesions‡	
PR	50 to <100% clearance of skin disease‡ from baseline without advancement in stage. May designate subset of Very Good PR based on 90 to <100% clearing of total body involvement	Without new tumors (T3) in MF/SS patients with T_1, T_2, or T_4
SD	<25% increase or <50% clearance in skin disease from baseline‡	Without new tumors (T3) in MF/SS patients with T_1, T_2, or T_4
PD†	1. ≥25% increase in skin disease from baseline‡ 2. Loss of response: in those with CR or PR, increase of skin score of greater than the sum of nadir plus 50% baseline score	New tumors (T_3) in MF/SS patients with T_1, T_2, or T_4 only skin disease

*A biopsy of normal appearing skin is unnecessary to assign a CR. However, a skin biopsy should be performed of a representative area of the skin if there is any question of residual disease (persistent erythema or pigmentary change) where otherwise a CR would exist. If histologic changes are suspicious or suggestive of PCL, the response should be considered a PR only.
†Whichever criterion occurs first.
‡One form of assessment of skin disease should be used throughout the trial. For a global response score and a designation of Very Good PR, a comparison of total body skin assessment based on mSWAT assessment at baseline is necessary. Regional or lesional skin scoring may also have CR, PR, SD and PD response but may not be representative of the response of skin disease on the entire body skin surface and cannot be used to assess global response.

8.1.2. Response in Nodes

ISCL/USCLC/EORTC response in lymph nodes

Response	Lugano PET-CT–based response	Lugano CT-based response	ISCL/EORTC/USCCL
CR	Complete metabolic response. Score 1, 2, or 3* with or without a residual mass on 5PS.	Target LNs†/nodal masses must regress to ≤1.5 cm LDi	All target LNs or nodal masses that previously were >1.5 cm are now ≤1.5 cm LDi by method used to assess size of LNs at baseline/screening or biopsy negative for lymphoma

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

PR	Partial metabolic response. Score of 4 or 5* with reduced uptake compared with baseline.	≥50% decrease in SPD of up to 6 target measurable LNs. No clear increase in nonmeasured LNs or new LN >1.5 cm LDi.	Cumulative reduction >50% of the SPD of up to 6 target LNs and no new LN >1.5 cm LDi unless proven pathologically negative for lymphoma
SD	No metabolic response. Score of 4 or 5* with no significant change in FDG uptake from baseline.	<50% decrease from screening/baseline in SPD of up to 6 target measurable LNs. Criteria for PD not met.	Fails to meet criteria for CR, PR or PD
PD	Progressive metabolic disease. Score of 4 or 5* with an increase in intensity of uptake.	1. Any LN of LDi >1.5 cm which has increased by ≥50% from PPD nadir 2. New LN >1.5 cm any axis 3. New or clear progression of preexisting nonmeasured LNs	1. Any LN >1.5 cm LDi which has increased by ≥50% from PPD nadir 2. Any prior LN <1.5 cm LDi, which has increased by >50% from PPD nadir to >1.5 cm LDi

FDG, fluorodeoxyglucose; LDi, longest diameter; LN, lymph node; 5PS, 5-point scale; PD, progressive disease; SDi, short axis (longest perpendicular diameter to the LDi); SPD, sum of the products of the perpendicular diameters for multiple lesions

*5PS: 1 = no FDG uptake > background; 2 = FDG uptake ≤ mediastinum; 3 = FDG uptake > mediastinum but ≤ liver; 4 = FDG uptake moderately > liver; 5 = FDG uptake markedly > liver and/or new lesions.
†Target LNs are those >1.5 cm with representative abnormal node positive pathologically for lymphoma. In MF/SS, this is currently the LN classification of N_3.

8.1.3. Response in Viscera

ISCL/USCLC/EORTC response in viscera

Response	Lugano PET-CT–based response	Lugano CT-based response
CR	Complete metabolic response. Score of 1, 2, or 3* with or without a residual mass on 5PS. No evidence of FDG-avid disease.	No extralymphatic sites of disease. Any abnormal size of organ at screening/baseline has returned to normal size. BM normal by morphology.
PR	Partial metabolic response. Score of 4 or 5* with reduced uptake compared with baseline and residual mass(es) of any size. Residual uptake in BM higher than normal but less than baseline.	1. ≥50% decrease in SPD from baseline of any measurable extranodal site 2. Spleen >50% regression in length beyond normal (≤13 cm) 3. No new lesions 4. No increase in nonmeasured lesions
SD	No metabolic response. Score of 4 or 5* with no significant change in FDG uptake from baseline. BM no change from BL.	Fails to attain criteria for CR, PR, or PD. No clear progression or improvement.
PD	1. Progressive metabolic disease 2. New FDG-avid foci consistent with lymphoma	1. New extranodal site >1 cm any axis or if <1 cm, must be attributable to lymphoma 2. An increase in LDi or SDi from nadir of 0.5 cm for lesions ≤2 cm or 1 cm for lesions >2 cm 3. Regrowth of previously resolved lesions 4. In the setting of splenomegaly at BL, an increase in splenic length by >50% from BL or if no splenomegaly at BL, new increase length >2 cm from BL 5. New or clear progression of preexisting non measured lesions

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

8.1.4. Response in Blood

Flow cytometry should capture absolute counts of CD4+CD7-, CD4+CD26- or another aberrant phenotype where:

- B0: absolute counts of either CD4+CD7- or CD4+CD26- are <250/µL cells
- B1: CD4+CD7- or CD4+CD26- counts are between 250/µL - <1,000/µL cells
- B2: CD4+CD7- cells or CD4+CD26- cells or other aberrant phenotypes* are ≥1,000/µL in the presence of a relevant T-cell clone in blood.

Most frequent aberrancies are loss of CD7 or CD26 but malignant cells can loss or express multiple markers. Examples of aberrant phenotypes are CD4+/CD3+ clones in presence of CD4/CD8 ratio ≥10 or CD4+/CD8+ cells.

The table below gives the response criteria for blood (modified from J. Scarisbrick et. al., 2018 / Olsen et. al., Blood 2021)

Response in Blood

Response	Definition
CR	B_0*, †
PR	>50% decrease in quantitative measurements of blood tumor burden from baseline in those with B_2 classification*, †, ‡
SD	Fails to attain criteria for CR, PR, or PD
PD§	• B_0 to B_2*, † or • >50% increase from baseline and ≥5000 neoplastic cells/µL or • Loss of response in those with PR who were originally B_2 at baseline, >50% increase from nadir and ≥5000 neoplastic cells/µl

*As determined by absolute numbers of neoplastic cells/mL by flow cytometry.

†The absolute number of $CD4^+CD26^-$ and/or $CD4+CD7^-$ lymphocytes may be used to assess blood involvement in clinical trials. In the case where more than one aberrant population of lymphocytes is recorded, the population with the highest absolute number at baseline should determine the B classification and the highest absolute number at each assessment should be used to determine the number of aberrant lymphocytes for response purposes.

‡There is no PR in those with B_1 disease at baseline as the difference within the range of neoplastic cells that defines B_1 is not considered significant and should not affect determination of global objective response.

§Whichever occurs first.

8.1.5. Global Response Score

Definitions of response in skin, lymph nodes, viscera and blood are incorporated to define the global response score (modified from E. A. Olsen et al., 2021) [4].

*It is recommended that not only the proportion of patients who achieve a response or an unfavorable outcome be calculated but a life table account for the length of the interval during which each patient is under observation also begenerated.

ISCL/USCLC/EORTC global disease response score in primary cutaneous lymphomas (modified from E. A. Olsen et al., 2021)

Global score*	Definition	Skin	Lymph nodes	Viscera	Blood
CR	Complete disappearance of all clinical evidence of disease	CR	All categories involved at baseline have CR/NI		

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

PR	Partial regression of measurable disease	CR	All categories involved do not have a CR/NI and no category has a PD
		PR	No category has a PD and if any category involved at baseline, at least one has a PR or CR
SD	Failure to attain CR, PR or PD representative of all disease	PR	No category has a PD and if any category involved at baseline, no CR or PR if any
		SD	CR/NI, PR, SD in any category and no category has a PD
PD	Progressive disease	PD in any category	

This table assumes that (1) all patients at baseline have measurable skin disease and (2) in patients with CTCL and no extracutaneous disease at baseline, any new nodal or visceral involvement constitutes PD in those compartments.

NI, noninvolved; SD, stable disease.

*This assumes that the response (CR, PR, SD, PD, or relapse) has been maintained for at least 4 weeks in any involved category.

All patients included in the study will be assessed for global response at 3 and 6 months even if there is a major protocol treatment deviation or if they are ineligible.

Each patient will be assigned one of the following categories: complete response (CR), partial response (PR), stable disease (SD), progressive disease (PD), relapse, early death or not evaluable.

To be assigned a status of CR or PR, the response must be confirmed by an assessment performed 3-6 weeks after the criteria for response are first met.

The status of PD does not need to be confirmed. However, in order to avoid premature protocol withdrawal, if the PD is not deemed clinically relevant, then patients are allowed to continue treatment until PD is confirmed.

Patients for whom response is not confirmed will be classified as "not evaluable", unless they meet the criteria for stable disease (or the criteria for partial response in case of an unconfirmed complete response).

Patients' response will also be classified as "not evaluable" if insufficient data were collected to allow evaluation per these criteria.

Early death is defined as any death occurring before the first per protocol time point of tumor re-evaluation.

8.2. Endpoint definition

8.2.1. Overall Response rate

The co-primary endpoint of the study is the overall response rate (ORR) defined as the proportion of patients who achieved a CR or PR, defined according to the EORTC-ISCL-USCLC criteria, as best result up to maximum 6 months after start of DT-7012.

8.2.2. Disease control rate

Disease control rate will be defined as the rate of patients with complete, partial response or stable disease, defined according to the EORTC-ISCL-USCLC criteria, as best result up to maximum 6 months after start of DT-7012.

8.2.3. Progression Free Survival Rate

Progression-free survival will be defined as a composite time-to-event variable, based on the locally assessed disease evaluation performed 3 and 6 months after start of DT-7012. Patients will be considered as failure if:

- they have progressed in any compartment or
- have confirmed unknown progression status
- have died at or before evaluation (all-cause death).

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Progression-free survival (PFS) is estimated from the time from start of DT-7012 to the first date of progressive disease, relapse or death from any cause. For patients who did not progress or die within their follow-up window, this endpoint will be censored on the date of last disease assessment.

8.2.4. Duration of response

Response duration will be measured from the time measurement criteria for CR/PR (whichever is first recorded) are first met until the first date that recurrent or progressive disease is objectively documented.

8.2.5. Time to next treatment

Time to next treatment is defined as the time from initiation of DT-7012 until the time the initiation of any total skin-equivalent treatment (phototherapy, TSEB) or systemic treatment is recorded. Maintenance treatments after documented CR or PR, and allogeneic HSCT will not be considered as TTNT events. For patients who did not have therecord of next treatment, this endpoint will be censored on the date of last adequate follow-up assessment.

9. SPECIFIC STUDY COMMITTEES

9.1. Steering Committee

This steering committee is made of the coordinating investigator (Adele de Masson), scientific director (Marie Beylot-Barry), drug development expert (a representative of Domain Therapeutics), biostatistician, sponsor representatives appointed for this study (URC and Promotion project managers). Principal investigators from recruiting sites will be invited to participate at every meeting. Ad hoc expert may be invited as needs arise.

Non-exhaustive list of data presented at the meeting: progress report (site activation, inclusion rate), number of patients currently under treatment, number of patients having completed the 3 months follow-up for ORR secondary criteria. If the meeting follows a DSMB meeting, the conclusion of the DSMB will be shared.

Roles: Define the overall structure of the study, coordinate information, determine the initial methodology and oversee the study.
Propose procedures to be followed during the study, acknowledging any recommendations from the Data Safety Monitoring Board, if applicable. The DRCI sponsor retains decision-making authority. The industrial (Domain therapeutics) or its representative is not a permanent member, only plays an advisory role and does not have any decision-making role in the steering committee.

9.2. Scientific Committee

The scientific committee is made of specialists in the condition and treatments under investigation, biostatisticians, methodologists. This is an off-shoot of the Steering Committee to which special competencies are added.
Committee members: Adele de Masson, Martine Bagot, Marie Beylot-Barry, statistician (Lucie Biard).
Roles: to define the purpose, to draft the protocol, to suggest modifications to the protocol during the study.

10. SAFETY ASSESSMENT - RISKS AND BURDEN ADDED BY THE STUDY

10.1. Description of Safety endpoints assessment parameters

All adverse events will be recorded. The investigator will assess whether those events are drug related (reasonable possibility, no reasonable possibility) and this assessment will be recorded in the database for all adverse events.

This study will use the International Common Terminology Criteria for Adverse Events (CTCAE), version 5.0, for adverse event reporting. A copy of the CTCAE can be accessed from the EORTC home page www.eortc.org\investigators-area\ctc.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

10.2. Anticipated methods and timetable for measuring, collecting and analysing the safety endpoints

The collection period will start from registration until 7 days after last dose of DT-7012 given as protocol treatment. Phone contacts are acceptable.

All adverse events must be followed until baseline or grade 1. If at the time of end of treatment patients had severe toxicities (AEs Grade 3 or Grade 4), follow up should be provided for these AEsuntil recovery to Grade 1 or baseline.

10.3. Recording and reporting adverse events

10.3.1. Definitions

- **Adverse event (AE)**

Any untoward medical occurrence in a subject, to whom a medicinal product is administered and which does not necessarily have a causal relationship with this treatment.

- **Serious adverse event (SAE)**

Any untoward medical occurrence that at any dose requires inpatient hospitalisation or prolongation of existing hospitalisation, results in persistent or significant disability or incapacity, results in a congenital anomaly or birth defect, is life-threatening, or results in death.
Certain medical events may jeopardise the subject or may require an intervention to prevent a SAE, known as "important medical events" should also be considered as serious adverse events.

- **Unexpected serious adverse reaction (SUSAR)**

A serious adverse reaction, the nature, severity or outcome of which is not consistent with the reference safety information.

- **Unexpected event**

Unexpected events affect the benefit-risk balance of the clinical trial, but are not suspected unexpected serious adverse reactions as referred to in Article 42. That notification shall be made without undue delay but no later than 15 days from the date the sponsor became aware of this event.

- **Urgent safety measure**

Where an unexpected event is likely to seriously affect the benefit-risk balance, the sponsor and the investigator shall take appropriate urgent safety measures to protect the subjects.
The sponsor shall notify the Member States concerned, through the EU portal, of the event and the measures taken.
That notification shall be made without undue delay but no later than seven days from the date the measures have been taken.

10.3.2. The role of the investigator

The investigator must **assess the seriousness of each adverse event** and record all serious and non-serious adverse events in the case report form (eCRF).

For all serious adverse events, the investigator must also notify the SAE to the safety department as soon as possible, except SAE not requiring a notification without delay described in section 10.3.2.2.2. “Serious adverse events that do not require the investigator to notify the sponsor without delay”

The investigator must **document** serious adverse events **as thoroughly as possible** and provide a definitive medical diagnosis, if possible.

The investigator must **assess the intensity** of the adverse events by using a rating scale for adverse events (CTCAE v5.0).

The investigator must **assess the causal relationship** between serious adverse events and:
- the investigational medicinal product (IMP).

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

- and/or interventions/procedures added by the study
- and/or auxiliary medicinal products and/or interaction between auxiliary medicinal product and the investigational medicinal product(s).

Three situations are possible for auxiliary medicinal products:
1 - In case of a suspected interaction between authorized auxiliary medicinal product and the investigational medicinal product(s), the reporting rules for the IMP apply: SAE must be notified to the sponsor's safety department as soon as possible.
2 - In case of adverse reactions related to non-authorized auxiliary medicinal products, the reporting rules for the IMP apply: SAE must be notified to the sponsor's safety department as soon as possible.
3 - In case of adverse reactions only related to authorized auxiliary medicinal products (without interaction with IMP): SAE must be notified to the relevant regional pharmacovigilance center (Centre Régional de Pharmacovigilance (CRPV)).

The method used by the investigator is based on 2 causality terms (EVCTM method):
- Related
- Not related

10.3.2.1 Serious adverse events that require the investigator to notify the sponsor without delay

The investigator notifies the sponsor without undue delay but not later than within 24 hours on the day the investigator becomes aware of any serious adverse event which occurs during a study that meets the description in article 41 of Regulation (EU) N°536/2014, with the exception of any event which is listed in the protocol and, if applicable, in the investigator's brochure as not requiring notification.

A serious adverse event is any untoward medical occurrence that:

1. results in death
2. is life-threatening to the participant enrolled in the study
3. requires hospitalisation or prolongation of existing hospitalisation
4. results in persistent or significant disability/incapacity
5. is a congenital anomaly/birth defect
6. important medical event*

* Certain medical events may jeopardise the subject or may require an intervention to prevent a SAE, known as "important medical events" should also be considered as serious adverse events.

Moreover, special circumstances require the investigator to notify the sponsor without delay are:
- medication errors,
- in utero exposure / pregnancy / breastfeeding
- uses outside what is foreseen in the protocol, including misuse and abuse of the product.

All pregnancies must be reported to the safety department by the investigators within 24 hours on the day the investigator becomes aware of it.

Special circumstances associated to an SAE must be reported to the safety department by the investigators.
Special circumstances not associated to an SAE must be reported to the CRF by the investigators and notify as a deviation to the protocol by the clinical trial unit.

10.3.2.2 Specific features of the protocol

10.3.2.2.1 Other events that require the investigator to notify without delay the sponsor

- ***Adverse events particularly followed by the sponsor for the safety and assessment***

These events must be notified by the investigator to the sponsor without delay on the day the investigator becomes aware of these adverse events, in the same manner and within the same deadline as for serious adverse events (see above).

For these adverse events particularly followed by the sponsor, please, check the "important medical event" box as a seriousness criterion in the SAE notification form.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

These adverse events particularly followed are the following:
- any DLT
- Infusion related reaction (grade ≥ 3)
- **New onset of auto-immune disorder (all grade)**
- ALAT / ASAT or Alcaline phosphatase increase grade ≥ 3
- Disease progression

- ***In utero* exposure**

The investigator must notify the sponsor without delay on the day the investigator becomes aware of any pregnancy that occurs during the study, even if it is not associated with an adverse event.
If the investigational medicinal product is genotoxic, every case of maternal or paternal exposure must be notified.

- *Exposure **while breastfeeding***

Exposure while breastfeeding occurs if an infant or a child may have been exposed to a medicinal product *via* the breast milk of a mother being treated with an investigational medicinal product.
Even if it is not associated with an adverse event, the investigator must notify the sponsor without delay on the day the investigator becomes aware of the exposure while breastfeeding.

10.3.2.2.2 Serious adverse events that do not require the investigator to notify the sponsor without delay

These serious adverse events are only recorded in the case report forms.
An extraction of all these SAE from the e-CRF will be carried out by Clinical Trial Unit (including randomization group if applicable, seriousness criteria, investigator's assessment and SAE grade if applicable) and sent by email to the Safety Department, every 3 months and at least once per year during the preparation of annual safety report. These extractions will be sent to the DRCI safety department at the following address: expertisecsi.drc@aphp.fr

- *Normal and natural course of the condition*:
  - planned hospitalisation for monitoring the CTCL [no deterioration in the participant's condition compared to baseline],
  - hospitalisation for routine treatment or for monitoring of the CTCL, not associated with a deterioration in the participant's condition,
  - emergency hospitalisation at inclusion or prolongation of hospitalisation after inclusion for monitoring the CTCL

- *Special circumstances*
  - Hospitalisation for a pre-existing illness or condition
  - Hospitalisation for a medical or surgical treatment scheduled prior to the study
  - Admission for social or administrative reasons

- *Adverse events related to treatments prescribed as part of the care provided during the study follow-up*

The investigator must report these adverse events to the relevant regional pharmacovigilance centre, (Centre Régional de Pharmacovigilance (CRPV) in France).

10.3.2.3 Period during which SAEs must be notified without delay by the investigator to the sponsor

The investigator must notify the sponsor without delay of any serious adverse events as defined in the corresponding section:
- from the date on which the participant signed the consent form.
- throughout the whole follow-up period required for the trial
- indefinitely, if the SAE is likely to be due to the investigational medicinal product or to the study interventions (e.g., serious reactions that could appear long after exposure to the medication, such as cancers or congenital abnormalities) *

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

*NB: In this last case, the investigator does not have to collect indefinitely in the case report form (CRF or eCRF) all SAEs possibly related to the clinical trial, but must notify them, to the sponsor, as soon as he/she becomes aware of them, by the SAE notification form or by email or by fax (as described below).

10.3.2.4 Procedures and deadlines for notifying the sponsor

The initial notification of an SAE must be provided in a written report signed by the investigator using a SAE notification form specific to the study and intended for this purpose (in the case report form).

Each item in this form must be completed by the investigator so that the sponsor can carry out the appropriate analysis.

The initial notification to the sponsor of a serious adverse event must be quickly followed by an additional detailed written report (or reports) so that the case outcome may be monitored by the Safety Department or to provide further information.

Whenever possible, the investigator will provide the sponsor with any documents that may be useful (medical reports, laboratory test results, results from additional examinations, etc.). These documents must be non-identifying. In addition, the documents must include the following: study acronym, number.

Any adverse event will be monitored until fully resolved (stabilisation at a level considered acceptable by the investigator, or return to the previous state) even if the participant has left the study.

The initial notification, the SAE follow-up reports and all other documents must be sent to the sponsor's Safety Department by email (eig-vigilance.drc@aphp.fr). It should be noted that it is possible to send SAE reports to the Safety Department by fax to +33 (0)1 44 84 17 99 only in the event of a failed attempt to send the SAE report by email (in order to avoid duplication).

When sending the email, please:
- Adopt a standardized nomming of the email subject in the following form:
<< Objet : YYYYYY_XXXXXX_jjmmaaaa (avec YYYYYY : code de la recherche, XXXXXX : acronyme de la recherche et jjmmaaaa : date de transmission). >>

- Send a SAE initial notification form and/or a follow-up report concerning a single participant for a given SAE, attachment may contain one (or more) document(s) (follow-up and hospitalization reports, for example).
The total size of the email must be less than 8 MB, otherwise please send several e-mails.

- Ensure that all documents transmitted (e.g., hospital reports) are anonymized and identified with the participant's identification number

For studies which use e-CRFs
- the investigator completes the SAE notification form in the e-CRF, then validates, prints, and signs the form before sending it by email;
- if it is not possible to connect to the e-CRF, the investigator should complete, sign, and send the SAE notification form to the Safety Department. As soon as the connection is restored, the SAE notification form in the e-CRF must be duly completed.

The investigator must respond to all requests from the sponsor for additional information (this new information must be also recorded in the e-CRF).

For all questions relating to the notification of an adverse event, the Safety Department can be contacted via email: vigilance.drc@aphp.fr.

For cases of *in utero* exposure, the investigator will complete the Notification and Follow-up form for a pregnancy occurring during participation in a study".
The investigator must monitor the pregnant woman throughout her pregnancy or until the pregnancy is terminated, and must notify the sponsor of the outcome of the pregnancy using this form.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.

If the outcome of the pregnancy falls within the definition of a serious adverse event (miscarriage, pregnancy termination, foetal death, congenital abnormality, etc.), the investigator must follow the procedure for reporting SAEs.
The initial pregnancy notification, the SAE follow-up reports, and any other documents will be sent to the sponsor according to the same procedures specified herein.
If the investigational medicinal product is genotoxic and if it was the father who was exposed, the investigator must obtain the pregnant woman's permission before collecting information about the pregnancy.

Specific case of studies conducted with the administration of a radioactive product (e.g., PET): in case of the onset of secondary cancer/cancer or development of a hereditary deficiency linked to exposure to ionising radiation, the investigator completes the specific form for cancers

10.3.3. Role of the sponsor

The sponsor, represented by its Safety Department shall continuously, throughout the trial, assess the safety of each investigational medicinal product throughout the study.

10.3.3.1 Analysis and declaration of serious adverse events

The sponsor assesses:

- the **seriousness** of all the adverse events reported
- the **causal relationship** between these events and the investigational medicinal product and/or interaction between the investigational medicinal products and the auxiliary medicinal products and/or specific interventions/procedures/examinations added by the study and any concomitant treatments,

All serious adverse events which the investigator and/or the sponsor reasonably believe may have a causal relationship with the investigational medicinal product are classed as suspected serious adverse reactions.

- the **expected or unexpected nature** of the serious adverse reactions

Any serious adverse reaction whose nature, severity, frequency or outcome is inconsistent with the safety information described in the summary of product characteristics, or in the investigator's brochure if the product is not authorised, is considered unexpected.
The sponsor, represented by its Safety Department, assesses the expected/unexpected nature of a serious adverse reaction based on the information described below**.**

❖ For serious adverse events likely to be related to the investigational medicinal product:
- refer to the section “Reference safety Information” of the Investigator's Brochure enclosed in CTIS platform and in the investigator study file

❖ For serious adverse events that may be related to the auxiliary medicinal product:
- refer to the SmPC for the medicinal product.
- Where the auxiliary medicinal product is authorised in the Member State concerned, no additional information apart from a valid SmPC is required.
- In principle, only authorised medicinal products should be used as auxiliary medicinal products in clinical trials (article 59 of the Regulation (EU) No 536/2014). However, in certain circumstances unauthorised auxiliary medicines may be used. This has to be justified in the protocol. The acceptable reasons for admitting non-authorised auxiliary medicinal products would be related to the availability of authorised auxiliary medicinal products (e.g., no authorised medicinal products exist in the EU, or the amounts available are not sufficient to satisfy the need of the clinical trial). The lower price of non-authorised auxiliary medicinal product shall not be considered as a legitimate justification.

❖ The serious adverse events potentially related to the interventions, procedures or examinations specific to the study are listed in the section “Benefit/risk assessment”.

Specific case of studies involving PET:

Exposure to ionising radiation can cause cancer or the development of hereditary deficiencies. The effective dose is 8 mSV, and so these adverse events are unlikely to occur.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

The sponsor will report all suspected unexpected serious adverse reactions (SUSARs) via Eudravigilance, within the regulatory time frame, to the competent authority:

- in the case of fatal or life-threatening suspected unexpected serious adverse reactions, as soon as possible and in any event not later than **seven days** after the sponsor became aware of the reaction;
- in the case of non-fatal or non-life-threatening suspected unexpected serious adverse reactions, not later than **15 days** after the sponsor became aware of the reaction;
- in the case of a suspected unexpected serious adverse reaction which was initially considered to be non-fatal or nonlife threatening but which turns out to be fatal or life-threatening, as soon as possible and in any event not later than seven days after the sponsor became aware of the reaction being fatal or life-threatening
- in the uncompleted case of fatal or life-threatening suspected unexpected serious adverse reactions, all additional information to complete this case have to be reported as soon as possible and in any event not later than eight days after the initial report
- All additional, relevant information must be declared by the sponsor in the form of follow-up reports within a period of 15 days starting from when the sponsor had this information.

The sponsor must notify all the investigators involved about any information that could adversely affect the safety of the research participants.

10.3.3.2 Analysis and declaration of other safety data

This means any new information that prompts a reassessment of the risk/benefit ratio of the trial or of the product under investigation, or which could be sufficient to consider changes to the use of the product, the trial procedures or trial documentation, or the suspension, cancellation or amendment of the research trial or of other similar trials.

The sponsor will report in CTIS platform without delay upon knowledge of any emerging safety issue and, if applicable, describe what measures have been taken.

Following the initial declaration of any emerging safety issue, the sponsor will address to competent authorities any additional relevant information about the emerging safety issue in the form of a follow-up report, which must be sent no later than 8 days after learning of the information.

10.3.3.3 Annual safety report

Once a year for the duration of the clinical study, the sponsor must produce an annual safety report (Development Safety Update Report - DSUR) which includes, in particular:

- a safety analysis for the research participants,
- a description of the patients included in the study (demographic profile etc.)
- a list of all the suspected serious adverse reactions that occurred during the period covered by the report,
- summary tables of all the serious adverse events that have occurred since the start of the study.

The sponsor produces one annual safety report (Development Safety Update Report - DSUR) for one clinical trial.

The report must be submitted in CTIS no later than 60 days after the anniversary of the date on which the competent authority authorised the trial.

The final annual safety report must be submitted in CTIS no later than 60 days after the end date of the clinical trial.
The end date of the clinical trial is defined as the date of the last visit of the last subject.

10.3.4. Data Safety Monitoring Board (DSMB)

The Data Safety Monitoring Board (DSMB) will be established by the sponsor. Its primary mission is to monitor safety data. It can have other missions, such as monitoring efficacy data (especially if the protocol includes interim analyses).
The sponsor is responsible for justifying the creation or absence of such a committee to the Competent Authority and to the Ethics Committee).

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

The DSMB must hold its first meeting before the first participant is enrolled and ideally before the protocol is submitted to the competent authority and the Ethics Committee).
All missions as well as the specific operating procedures of the DSMB are described in the study's DSMB charter. The DSMB has no role in deciding dose escalation in the context of this TITE-CRM design.

The DSMB members are: Dr. Florent Amatore (dermatologist, CTCL expert, CHU Timone), Pr. Florent Grange (oncodermatologist, CTCL expert, CH de Valence), Dr. Saskia Oro (dermatologist, CTCL expert, CHU Henri-Mondor). The independent statistician is Dr. Moreno Ursino (Biostatistician, Methodologist early phase trials designs expert, INSERM-INRIA HeKa, Paris Santé Campus).
The DSMB will operate in accordance with the sponsor's procedures. The DSMB works in an advisory capacity only and the sponsor retains all decision-making authority.

11. DATA MANAGEMENT

11.1. Data collection procedures

All source data will be entered in the eCRF.

11.2. Identification of data recorded directly in the CRFs which will be considered as source data

None

11.3. Right to access data and source documents

11.3.1. Data access

In accordance with GCPs and appendix 1 of the European Regulation N°536-2014:

- the sponsor is responsible for ensuring all parties involved in the study agree to guarantee direct access to all locations where the study will be carried out, the source data, the source documents and the reports, for the purposes of the sponsor's quality control and audit procedures or inspections by the competent authority
- the sponsor declares that investigators and participating institution will ensure the persons in charge of monitoring, quality control and auditing or inspecting the clinical trial have access to the documents and personal data strictly necessary for these tasks, in accordance with the statutory and regulatory provisions in force

11.3.2. Source documents

Source documents are defined as any original document or item that can prove the existence or accuracy of a data or a fact recorded during the study. These documents will be kept in accordance with the regulations in force by the investigator or by the hospital in the case of a hospital medical file.

The type of source documents in the present study are the medical file, original biological examination results, summary from imaging examinations, skin biopsies reports, and clinical photographs in any.

11.3.3. Data confidentiality

The persons responsible for the quality control of clinical studies will take all necessary precautions to ensure the confidentiality of information relating to the investigational medicinal products, the study, the study participants and in particular their identity and the results obtained.
These persons, as well as the investigators themselves, are bound by professional secrecy.
During and after the clinical trial, all data collected concerning the participants and sent to the sponsor by the investigators (or any other specialised collaborators) will be rendered non-identifying.
Under no circumstances shall the names and addresses of the participants involved be shown.
The sponsor will ensure that each participant has given written permission for any personal information about him or her which is strictly necessary for the quality control of the study to be accessed.

11.4. Data processing and storage of research documents and data

11.4.1. Identification of the data processing manager and location(s)

Clinical research assistants under the supervision of the investigators will be responsible for data entry.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Statistical analysis will be performed by URC Saint-Louis (AP-HP, Paris, France).

11.4.2. Data entry

Non-identifying data will be entered electronically via a web browser (Cleanweb).

11.5. Data ownership

AP-HP is the owner of the data. The data cannot be used or disclosed to a third party without its prior permission.

12. STATISTICAL ASPECTS

12.1. Description of statistical methods to be used including the timetable for the planned interim analyses

Patients' characteristics overall and by dose level [Yap C et al. Enhancing quality and impact of early phase dose-finding clinical trial protocols: SPIRIT Dose-finding Extension (SPIRIT-DEFINE) guidance BMJ 2023;383:e076386; Yap C et al. Enhancing reporting quality and impact of early phase dose-finding clinical trials: CONSORT Dose-finding Extension (CONSORT-DEFINE) guidance BMJ 2023;383:e076387], as follows: qualitative variables with count and percent, continuous variables with median and interquartile range.

Primary endpoint
The continual reassessment method (CRM) is a model-based dose-escalation design for the phase I clinical trials, using the sequential estimation of the DLT rate at each dose level, relying on a parametric model for the dose-toxicity relationship which is sequentially updated based on accumulated DLT data during the trial, to estimated and recommend the maximum tolerated dose (MTD) [O'Quigley et al. Biometrics 1990; 46 (1): 33-48]. The CRM design has long been shown to outperform the 3+3 rule-based design in recommending the correct MTD and dosing more trial patients close to the MTD [O'Quigley J. Stat Med 1999; 18: 2683-90; Iasonos A et al. Clin Trials 2008; 5: 465–77; Wheeler et al. BMC Med Res Method 2019; 19: 18; Chiuzan C & Dehbi HM. Clin Trials 2024; 21 (3): 350-357]. The time-to-event extension of the CRM (TITE-CRM), relying on a weighted likelihood for the binomial toxicity events, allows for continuous accrual [Cheung YK & Chappell R. Biometrics 2000;56(4):1177-82].
The design will rely on the one-stage Bayesian TITE-CRM using the power model and linear weights, with four dose levels, cohorts of size 1 and a maximum sample size of 30 patients (see below section 12.2 and safety rule below). The initial dose will be dose level 2 (see section 2.6). The target DLT rate for the definition of the MTD will be 30% within the primary observation window of 3 weeks.
Following the CRM dose escalation design, the Bayesian dose-toxicity power model will be estimated sequentially after each cohort of patients has been included (cohort size=1). Exact timing of dose-assignment analyses will directly depend on patients' accrual and cannot be specified in advance, beside noting that an average rate of about 1.7 patients per month is anticipated (30 patients in 18 months). With the first cohort starting at dose level 2, each of the following cohorts will be assigned targeting the estimated MTD based on available DLT information at 3 weeks, accounting for the target DLT rate of 30% and the safety rules below [O'Quigley et al. Biometrics 1990; 46 (1): 33-48; Mozgunov P & Jaki T. Stat Med. 2019;39(7):906–922.]. Based on centrally collected DLT data (web-based eCRF and continuous monitoring), all analyses (dose-assignments and final) will be centrally performed at the Department of biostatistics in Saint Louis hospital. Dose-assignments will therefore be performed centrally, independently of the patient's physician following the CRM scheme and the safety rule detailed hereafter.
Safety rules will be implemented for the dose-finding procedure:
- No dose skipping will be allowed during dose escalation;
- For each newly escalated dose level, at least three patients must have been treated at this level prior to escalating to the next above dose level;
- For each newly escalated dose level, the first three patients treated at this dose level will be included at least one week apart;
- Dose escalation will be terminated before the planned maximum number of patients have been treated if the lower limit of the posterior 90% probability interval of the DLT rate at Dose Level 1 (DL1) exceeds the target DLT rate of 30%

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Following its charter, the DSMB will hold a meeting at the end of the primary window of the 3rd included patient.
The dose-escalation design will be calibrated to obtain desirable operating characteristics (notably, probability of correct MTD selection, probability of overdose selection, probability of DLT in trial participants) [O'Quigley et al. Biometrics 1990; 46 (1): 33-48; Mozgunov P & Jaki T. Stat Med. 2019;39(7):906–22]. Specifically, the dose skeleton will be defined using the indifference interval approach [Lee SM & Cheung YK. Clinical Trials; 6 (3): 227-38] and a least informative prior variance [Lee SM & Cheung YK. Stat Med 2011; 30 (17): 2081-89] will be used for the working model parameter. To that aim, a comprehensive simulation study will be reported in the statistical analysis plan, prior to the first inclusion.
Of note, at the end of the trial, secondary analyses of toxicity (DLT events) and ORR will account for repeated administrations of treatment and the effective dose administered at each cycle. Such analyses will be described in the statistical analysis plan.
Analyses and any data will be presented to the DSMB upon their request, as defined in the DSMB charter.

Secondary endpoints will be described overall and by dose level. Continuous variables will be described with the median and interquartile or minimum-maximum range; dichotomous and categorical variables will be described with counts and percentages; time-to-event variables with censoring will be described using Kaplan Meier estimator or the usual cumulative incidence estimator in case of competing risks.
Of note, safety endpoints will be monitored continuously during the trial and adverse event reporting will be performed as described in section 10.3.

12.2. Calculation hypotheses for the number of participants required and the results

In the targeted population (advanced-stage refractory and relapsed CTCL treated with an anti-CCR8 monoclonal antibody), we consider a maximum tolerated toxicity rate of 30 % (3 out of 10 patients) to guide dose finding.
Using the continual reassessment method, a maximum sample size of 30 patients corresponds to 60% average probability of correct selection of the MTD, given four candidate dose levels and a 1.8 odds ratio in between adjacent dose levels for calibration logistic dose-toxicity curves [Cheung YK. Clin Trials. 2013;10(6):852-861].
At the end of the trial, beside the MTD recommendation obtained with the CRM, adjacent dose levels will also be considered, in terms of safety and efficacy signals, at 3 weeks and after accounting for the repeated administration of treatment, for further dose optimization in subsequent clinical trials.

12.3. Anticipated level of statistical significance

Primary analyses will be performed using Bayesian inference, with a dose-finding objective, without using statistical hypotheses testing.
Otherwise, analyses of patient's characteristics and secondary endpoints will be of descriptive nature.
In the case of post-hoc comparisons with hypotheses testing performed, a 5% two-sided significance level will be used.

12.4. Statistical criteria for termination of the study.

A safety stopping rule will be implemented with regards to the estimated DLT rate during the trial. Dose escalation will be terminated before the planned maximum number of patients have been treated [Wages N & Petroni G. BMC Cancer 2018; 18: 133]:

- If the lower limit of the posterior 90% probability interval of the DLT rate at Dose Level 1 (DL1) exceeds the target DLT rate of 30% [Natarajan L & O'Quigley J. Stat Med. 2003; 22: 1829–36]

Otherwise, we will enroll a maximum number of 30 patients with a cohort size of 1 and safety rules for staggered inclusions.

12.5. Method for taking into account missing, unused or invalid data

Invalid or missing data will be described.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Patients with missing data on the primary endpoint (eg, patients lost to follow up) will be included in the primary analysis using the TITE-CRM framework to handle such incomplete observations [Biard L et al; J Biopharm Stat. 2021;31(2):156-167].
For the other analyses (patients' characteristics, secondary endpoints), complete case analysis will be performed primarily; sensitivity analyses may be performed using simple imputation (e.g. treatment failure in case of missing data on ORR).

12.6. Choice of individuals to be included in the analyses

- Given the safety primary endpoint, primary analyses (dose assignments and final dose finding analysis) will be performed on the sample of patients treated, in the dose level group corresponding to the dose level they were administered for the first cycle. Intercurrent events during the primary event window will be collected; incomplete observations for the primary endpoint due to intercurrent event will be included in the primary analysis, using the TITE design [Biard L et al; J Biopharm Stat. 2021;31(2):156-167].
- Otherwise, the following populations will be considered for the other outcomes: intent-to-treated (all included patients, in their assigned dose level group), modified ITT (all included patients in their assigned dose level group, except those, if any, who eventually did not receive the study treatment), as treated (included patients in the dose level group corresponding to the dose level they in fact received for the first cycle).

13. QUALITY CONTROL AND ASSURANCE

Every clinical trial sponsored by AP-HP is ranked according to the projected risk incurred by study participants using the classification system specific to AP-HP sponsored clinical trial.

13.1. General organisation

The sponsor must ensure the safety and respect of individuals who have agreed to participate in the study. The sponsor must implement a quality assurance system to best monitor the implementation of the study in the investigation centres.

For this purpose, the sponsor shall assign Clinical Research Associates (CRA) whose primary role is to carry out regular follow-up visits at the study locations, after having carried out the initial visits.
The purpose of monitoring the study, as defined in Good Clinical Practices (GCP section 5.18.1), is to verify that:

- the rights, safety and protection of the research participants are met
- the data reported are exact, complete and consistent with the source documents
- the study is carried out in accordance with the protocol in force, the GCPs and the statutory and regulatory provisions in force

13.1.1. Strategy for centre opening

The strategy for opening the centres established for this study is determined using the appropriate monitoring plan.

13.1.2. Scope of centre monitoring

In the case of this risk D study the appropriate monitoring level has been determined based on the complexity, the impact and the budget for the study. Therefore, in agreement with the coordinating investigator, the sponsor has determined the logistical score and impact, resulting in a study monitoring level to be implemented: level **D.**
These various levels are defined in the monitoring charter for research involving human participants

13.2. Quality control

A Clinical Research Associate (CRA) appointed by the sponsor will be responsible for the good completion of the study, for collecting, documenting, recording and reporting all handwritten data, in accordance with the Standard Operating Procedures applied within the DRCI (Clinical Research and

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Innovation Department) and in accordance with Good Clinical Practices as well as the statutory and regulatory requirements.

The investigator and the members of the investigator's team agree to make themselves available during regular Quality Control visits carried out by the Clinical Research Associate. During these visits, the following elements will be reviewed depending on the monitoring level:

- written consent
- compliance with the study protocol and with the procedures defined therein
- quality of the data collected in the case report form: accuracy, missing data, consistency of the data with the "source" documents (medical files, appointment books, original copies of laboratory results, etc.)
- management of the treatments used.

13.3. Case report forms

The case report forms should only contain the data needed to analyse the study and publish the results. All other data needed to monitor the participants during the study and for standard of care are recorded in the medical file.
All information required by the protocol must be entered in the case report forms. The data must be collected as and when they are obtained, and clearly recorded in these case report forms. Any missing data must be coded.
Every site will have access to the electronic case report forms via a web-based data collection system. Investigators will be given a document offering guidance on using this tool.
When the investigators complete the case report form via the Internet, the CRA can view the data quickly and remotely. The investigator is responsible for the accuracy, quality and relevance of all the data entered. In addition, the data are immediately verified as they are entered, thanks to consistency checks. To this end, the investigator must validate any changes to the values in the case report form. An audit trail will be kept of all changes. A justification can be added when applicable, as a comment.
The investigator must archive a copy of the authenticated document that was issued to the sponsor.

13.4. Management of non-compliances

Any events that occur as a result of non-compliance – by the investigator or any other individual involved in running the study – with the protocol, standard operating procedures, good clinical practices or statutory and regulatory requirements must be recorded in a declaration of non-compliance and sent to the sponsor.

These non-compliances will be managed in accordance with the sponsor's procedures.

The sponsor shall notify the Member States concerned about a serious breach of this Regulation or of the version of the protocol applicable at the time of the breach through the EU portal without undue delay but not later than seven days of becoming aware of that breach.

13.5. Audits/inspections

The investigators agree to consent to the quality assurance audits carried out by the sponsor as well as the inspections carried out by the competent authorities. All data, documents and reports may be subject to regulatory audits and inspections. These audits and inspections cannot be refused on the grounds of medical secrecy.

An audit can be carried out at any time by individuals appointed by the sponsor and independent of those responsible for the research. The aim of the audits is to ensure the quality of the study, the validity of the results and compliance with the legislation and regulations in force.

The persons who manage and monitor the study agree to comply with the sponsor's requirements and with the competent authority regarding study audits or inspections.

The audit may encompass all stages of the study, from the development of the protocol to the publication of the results, including the storage of the data used or produced as part of the study.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

13.6. Principal Investigator's commitment to assume responsibility

Before starting the study, each investigator will give the sponsor's representative a copy of his/her updated personal *curriculum vitæ*, signed and dated less than one year, with his/her RPPS number (*Répertoire Partagé des Professionnels de Santé*, Collective Database of Health Professionals, for France). The CV must include any previous involvement in clinical research and related training.
Any conditions, such as economic interests and institutional affiliations that might influence the impartiality of the investigators shall be presented.

Each investigator will commit to comply with legislation and to conduct the study in line with GCP, in accordance with the Declaration of Helsinki. Each investigator will give the sponsor's representative a GCP certificate dated fewer than three years ago.

The Principal Investigator at each participating site will sign a commitment of responsibility (standard DRCI document) which will be sent to the sponsor's representative.
The investigators and their staff will sign a delegation of duties form specifying each person's role and will provide their CVs.

13.7. Suitability of the facilities

Before starting the study, each clinical trial sites will give the sponsor's representative a duly justified written statement on the suitability adapted to the nature and use of the investigational medicinal product and including a description of the suitability of facilities, equipment, human resources and description of expertise.

14. ETHICAL AND LEGAL CONSIDERATIONS

14.1. Methods for informing research participants and obtaining their consent

According to article 29 of European regulation N°536/2014, No clinical trials on medicinal products for human use can be carried out on a person without his/her freely given and informed consent, obtained in writing after the person has been given the information specified in article 29.2 of the European Regulation.

A reflection period of **at least 24 hours** is given to the individual between the time when he or she is informed and when he or she signs the consent form.

The person's freely-given, written, informed consent will be obtained by the principal investigator or a physician representing the investigator before the person is enrolled in the study, after at least 24 hours of reflection and within one month prior to the baseline visit.

A copy of the information note and consent form, signed and dated by the research participant and by the principal investigator or the physician representing the investigator will be given to the individual prior to their participation in the study. The principal investigator or the physician representing him/her will keep a copy.

For French sites only: At the end of the study, one copy will be placed in a tamper-proof sealed envelope containing all the consent forms. This envelope will be archived by the sponsor.

In addition, the investigator will specify in the person's medical file the person's participation in the research, the procedures for obtaining his/her consent as well as the methods used for providing information for the purpose of collecting it. The investigator will retain one copy of the signed and dated consent form.

14.2. Prohibition from participating in another clinical study or exclusion period set after the study

During the study, the participant cannot participate in another study on a health product. The participant can however participle in any other interventional or non-interventional studies.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

After the end of the participation, there is an exclusion period of 6 months.

14.3. Compensation for participants

None**.** However, the transportation costs will be reimbursed to the patients for the visits added by the research.

14.4. Registration on the National Register of study participants to clinical trial concerning the products mentioned in Article L. 5311-1 of the Code de la santé publique

No registration will be performed because there is no requirement:
In France, in accordance with Article L.1121-16 of the Code de la Santé Publique (French the Public Health Code), registration is required:

- for healthy persons who volunteer to participate in a study;
- for patients who participate in a study unrelated to their illness or condition;
- in certain cases at the request of the Ethics Committee.

14.5. Authorisation for the research location

In France, the study will be carried out in treatment units on individuals presenting a clinical condition for which the units are specialised and who require interventions that are routinely performed at those units. The study requires the presence of a unit specializing in early phase clinical trials (Centres Labellisés essais de phase précoce).

14.6. Legal obligations

14.6.1. Role of the sponsor

Assistance Publique - Hôpitaux de Paris (AP-HP) is the sponsor of this study and, by delegation, the DRCI (Clinical Research and Innovation Department) carries out the study's missions in accordance with regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014 and all national Laws. Assistance Publique - Hôpitaux de Paris reserves the right to halt the study at any time for medical or administrative reasons. In this case, notification will be sent to the investigator.

14.6.2. Request for authorisation

Prior to starting the study, AP-HP, as sponsor, must obtain authorisation from the Competent Authority and the Research Committee for this clinical trial on medicinal product for human use within the scope of its authority and in accordance with in force legislation and regulatory requirements.

14.6.3. Procedures relating to data protection regulations

The computer file used for this research is implemented in accordance with French (amended “Informatique et Libertés” law governing data protection) and European (General Data Protection Regulation – GDPR) regulations.

<u>For France:</u>

- Commitment to comply with “Reference Methodology” MR-001

This research is governed by the CNIL (French Data Protection Agency) “Reference Methodology for processing personal data used within the scope of health research” (amended MR-001). AP-HP, as sponsor of the research, has signed a declaration of compliance with this “Reference Methodology”.

<u>For data processed outside French territory:</u>

- *Data managed in a European Union country:*

Data will be entered by local team from Germany and Spain on the eCRF.
The eCRF server is located in AP-HP, France.

- Data and biological samples for PK and ADA analysis:

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

These will be sent to 360°Biolabs for analysis. The subcontractor has laboratories in different locations. The samples may be sent to a laboratory based in the EU or based in Australia. In any case, a contract will be drawn up to ensure compliance of the transfer and subsequent processing. The patients will be informed of this data processing.

14.6.4. Start of the Clinical Trial

The sponsor shall notify each Member State concerned of the start of a clinical trial and the start of the recruitment through the EU portal. That notification shall be made within 15 days from the start of the clinical trial.

14.6.5. Amendments to the Clinical Trial

Any substantial modification to the protocol by the coordinating investigator must be sent to the sponsor for approval. After approval is given, the sponsor must obtain approval from the Ethics Committee) and authorisation from the competent authority within the scope of their respective authorities, before the amendment can be implemented.
The information note and the consent form can be revised if necessary, in particular in case of a substantial amendment to the study or if adverse reactions occur.

14.6.6. End of the Clinical Trial

The end of the Clinical Trial corresponds to the end of the participation of the last person who participates to the Clinical Investigation.
The end of the recruitment of subjects for the clinical trial and the end of the clinical trial should be notified in CTIS. That notification shall be made within 15 days from the end of the clinical trial in relation to that Member State.

14.6.7. Summary of the results of the clinical trial

According to article 37.4 of the European regulation n°536/2014, irrespective of the outcome of a clinical trial, within one year from the end of a clinical trial in all Member States concerned, the sponsor shall submit to the CTIS a summary of the results of the clinical trial. This summary shall be accompanied by a summary written in a manner that is understandable to laypersons.

14.6.8. Archiving

Specific documents for a clinical trial on medicinal product for human use will be archived by the investigator and the sponsor for 25 years after the end of the research.

This indexed archiving includes, in particular:

- A sealed envelope for the investigator containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- For French sites only: A sealed envelope for the sponsor, containing a copy of all the information notes and consent forms signed by all individuals at the centre who participated in the study;
- "Study" binders for the Investigator and the sponsor, including (non-exhaustive list):
  - the successive versions of the protocol (identified by the version number and its date), and any appendices
  - the competent authority authorisations and Research Ethics Committee decisions
  - any correspondence
  - the enrolment list or register
  - the appendices specific to the research
  - final study report
- The data collection documents.

15. FUNDING AND INSURANCE

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

15.1. Funding sources

This study is funded by the Agence Nationale de la Recherche (ANR / RHU SPRINT / WP 6).

15.2. Insurance

Pursuant to Article L.1121-10 of the Code de la Santé Publique (French Public Health Code), insurance policies must guarantee the civil liability of the sponsor and that of any contributor and cover pecuniary consequences of damages arising from the study involving human participants.

For the duration of the study, the Sponsor will take out an insurance policy covering the sponsor's own public liability, as well as the public liability for all the physicians involved in the study. The sponsor will also provide full compensation for any damages caused by the study to the participant enrolled and their beneficiaries, unless the sponsor can prove that the harm is not the fault of the sponsor or any collaborator. Compensation cannot be refused on the grounds of a third-party act or the voluntary withdrawal of the person who initially consented to participate in the study.

Assistance Publique-Hôpitaux de Paris (AP-HP) has taken out insurance with HDI-GLOBAL SE through the insurance broker BIOMEDIC-INSURE for the full study period, which covers its own public liability and that of any collaborator (physician or research staff), in accordance with Article 76(1) of regulation (EU) No 536/2014 of the European Parliament and of the council of 16 April 2014 and Article L.1121-10 of the *Code de la Santé Publique* (French Public Health Code).

16. PUBLICATION RULES

The author(s) of any publication relating to this study must include the AP-HP among their affiliations and must name the sponsor AP-HP (DRCI) and the source of funding, if funded by a call for tenders (e.g., national or regional PHRC); a copy of the publication must be sent to the sponsor (see below for rules governing affiliation, and naming of the sponsor and funders). All investigators who have enrolled patients must be authors of the publication, with an order of authorship depending on the number of patients enrolled in the clinical trial.

16.1. Mention of AP-HP affiliation for projects sponsored by AP-HP

- If an author has several affiliations, the order in which the institutions are mentioned (AP-HP, University, INSERM, etc.) is unimportant
- However, if the study is funded in the context of an internal AP-HP call for tender, the first affiliation must be "AP-HP"
- Each of these affiliations must be identified by an address and separated by a semicolon (;)
- The AP-HP institution must feature under the acronym "**AP-HP**" first in the address, specifically followed by: **AP-HP**, hospital, department, city, postcode, France

16.2. Mention of the sponsor AP-HP (DRCI) in the acknowledgements of the text

- "The sponsor was Assistance Publique – Hôpitaux de Paris (Direction de la Recherche Clinique et de l'Innovation)"

16.3. Mention of the financial backer in the acknowledgements of the text

This study is funded "Hors Appel d'offres" (outside the context of a call for proposal – the study is funded by RHU SPRINT): this financial support will be mentioned according to the contractual conditions.

This study has been registered on the website http://clinicaltrials.gov/ under number *(add the registration number when the study is registered)*.

17. BIBLIOGRAPHY

1. Willemze R, Cerroni L, Kempf W, Berti E, Facchetti F, Swerdlow SH, et al. The 2018 update of the WHO-EORTC classification for primary cutaneous lymphomas. Blood. 2019 18;133(16):1703–14.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

2. Dobos G, de Masson A, Ram-Wolff C, Beylot-Barry M, Pham-Ledard A, Ortonne N, et al. Epidemiological changes in cutaneous lymphomas: an analysis of 8593 patients from the French Cutaneous Lymphoma Registry. Br J Dermatol. 2020 Oct 31;
3. Scarisbrick JJ, Prince HM, Vermeer MH, Quaglino P, Horwitz S, Porcu P, et al. Cutaneous Lymphoma International Consortium Study of Outcome in Advanced Stages of Mycosis Fungoides and Sézary Syndrome: Effect of Specific Prognostic Markers on Survival and Development of a Prognostic Model. J Clin Oncol. 2015 Nov 10;33(32):3766–73.
4. Olsen EA, Whittaker S, Willemze R, Pinter-Brown L, Foss FM, Geskin LJ, et al. Primary Cutaneous Lymphoma: Recommendations for Clinical Trial Design and Staging Update from the ISCL, USCLC, and EORTC. Blood. 2021 Nov 10;blood.2021012057.
5. Latzka J, Assaf C, Bagot M, Cozzio A, Dummer R, Guenova E, et al. EORTC consensus recommendations for the treatment of mycosis fungoides/Sézary syndrome - Update 2023. Eur J Cancer. 2023 Sep 18;195:113343.
6. de Masson A, Beylot-Barry M, Ram-Wolff C, Méar J, Dalle S, d'Incan M, et al. Allogeneic transplantation in advanced cutaneous T-cell lymphomas: a propensity score matched controlled prospective trial. Lancet. 2023;
7. Giustiniani J, Dobos G, Moins-Teisserenc H, Eustaquio T, Battistella M, Ortonne N, et al. CCR8 is a new therapeutic target in cutaneous T-cell lymphomas. Blood Adv. 2022 Jun 14;6(11):3507–12.
8. McCully ML, Ladell K, Andrews R, Jones RE, Miners KL, Roger L, et al. CCR8 Expression Defines Tissue-Resident Memory T Cells in Human Skin. J Immunol. 2018 Mar 1;200(5):1639–50.
9. Watanabe R, Gehad A, Yang C, Scott LL, Teague JE, Schlapbach C, et al. Human skin is protected by four functionally and phenotypically discrete populations of resident and recirculating memory T cells. Sci Transl Med. 2015 Mar 18;7(279):279ra39.
10. Campbell JR, McDonald BR, Mesko PB, Siemers NO, Singh PB, Selby M, et al. Fc-Optimized Anti-CCR8 Antibody Depletes Regulatory T Cells in Human Tumor Models. Cancer Res. 2021 Jun 1;81(11):2983–94.
11. Kim YH, Bagot M, Pinter-Brown L, Rook AH, Porcu P, Horwitz SM, et al. Mogamulizumab versus vorinostat in previously treated cutaneous T-cell lymphoma (MAVORIC): an international, open-label, randomised, controlled phase 3 trial. Lancet Oncol. 2018 Sep;19(9):1192–204.
12. Bagot M, Porcu P, Marie-Cardine A, Battistella M, William BM, Vermeer M, et al. IPH4102, a first-in-class anti-KIR3DL2 monoclonal antibody, in patients with relapsed or refractory cutaneous T-cell lymphoma: an international, first-in-human, open-label, phase 1 trial. Lancet Oncol. 2019 Aug;20(8):1160–70.
13. Leupin N, Zinzani PL, Morschhauser F, Dalle S, Maerevoet M, Michot JM, et al. Cusatuzumab for treatment of CD70-positive relapsed or refractory cutaneous T-cell lymphoma. Cancer. 2022 Mar 1;128(5):1004–14.
14. Starodub A, Opyrchal M, Balaraman R, Barve R, Fel D, Li R, et al. Safety and clinical activity of a novel anti-CCR8 antibody (LM-108) as a single agent or in combination with pembrolizumab in patients with advanced solid tumors: Results of phase 1 study. In 2023.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

18. LIST OF ADDENDA

Each addendum and the log of addenda versions are attached, independently of the protocol. Each addendum can be modified (change of addendum version) without modifying the protocol version.

18.1. List of investigators

18.2. Serious Adverse Events notification form

18.3. Pregnancy notification form

18.4. SmPC or Investigator's Brochure

18.5. Questionnaire or scale

mSWAT tool[*]

Body area according to Lund and Browder	% BSA body area	Lesion subtype		
		% BSA patch	% BSA plaque/papule	% BSA tumor
Head	7			
Neck	2			
Left upper arm	4			
Left forearm	3			
Left hand	2.5			
Right upper arm	4			
Right forearm	3			
Right hand	2.5			
Anterior trunk	13			
Genitalia	1			
Posterior trunk	13			
Buttocks	5			
Left thigh	9.5			
Left leg	7			
Left foot	3.5			
Right thigh	9.5			
Right leg	7			
Right foot	3.5			
Total BSA	100			
Weighting factor		1	2	4
Product BSA × weighting factor				
mSWAT score				

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

*The mSWAT score assesses the total skin involvement in CTCL by the summation of the product of the approximate BSA covered by 1 of 3 types of skin lesions (patch, plaque, or tumor) multiplied by the weighting of 1, 2, or, 4, respectively, for lesion type. The percent BSAs per body area in the above table are from Lund and Browder.

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

Skindex-29 questionnaire

Skindex-29 (source questionnaire)

How often during the past week do these statements describe you?		Never	Rarely	Some-times	Often	All the time
1.	My skin hurts	☐	☐	☐	☐	☐
2.	My skin condition affects how well I sleep	☐	☐	☐	☐	☐
3.	I worry that my skin condition may be serious	☐	☐	☐	☐	☐
4.	My skin condition makes it hard to work or do hobbies	☐	☐	☐	☐	☐
5.	My skin condition affects my social life	☐	☐	☐	☐	☐
6.	My skin condition makes me feel depressed	☐	☐	☐	☐	☐
7.	My skin condition burns or stings	☐	☐	☐	☐	☐
8.	I tend to stay at home because of my skin condition	☐	☐	☐	☐	☐
9.	I worry about getting scars from my skin condition	☐	☐	☐	☐	☐
10.	My skin itches	☐	☐	☐	☐	☐
11.	My skin condition affects how close I can be with those I love	☐	☐	☐	☐	☐
12.	I am ashamed of my skin condition	☐	☐	☐	☐	☐
13.	I worry that my skin condition may get worse	☐	☐	☐	☐	☐
14.	I tend to do things by myself because of my skin condition	☐	☐	☐	☐	☐
15.	I am angry about my skin condition	☐	☐	☐	☐	☐
16.	Water bothers my skin condition (bathing, washing hands)	☐	☐	☐	☐	☐
17.	My skin condition makes showing affection difficult	☐	☐	☐	☐	☐
18.	I worry about side-effects from skin medications / treatments	☐	☐	☐	☐	☐
19.	My skin is irritated	☐	☐	☐	☐	☐
20.	My skin condition affects my interactions with others	☐	☐	☐	☐	☐
21.	I am embarrassed by my skin condition	☐	☐	☐	☐	☐
22.	My skin condition is a problem for the people I love	☐	☐	☐	☐	☐
23.	I am frustrated by my skin condition	☐	☐	☐	☐	☐
24.	My skin is sensitive	☐	☐	☐	☐	☐
25.	My skin condition affects my desire to be with people	☐	☐	☐	☐	☐
26.	I am humiliated by my skin condition	☐	☐	☐	☐	☐
27.	My skin condition bleeds	☐	☐	☐	☐	☐
28.	I am annoyed by my skin condition	☐	☐	☐	☐	☐
29.	My skin condition interferes with my sex life	☐	☐	☐	☐	☐
30.	My skin condition makes me tired	☐	☐	☐	☐	☐

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

ENGLISH

EORTC QLQ-C30 (version 3)

We are interested in some things about you and your health. Please answer all of the questions yourself by circling the number that best applies to you. There are no "right" or "wrong" answers. The information that you provide will remain strictly confidential.

Please fill in your initials: ⌊⌊⌊⌊

Your birthdate (Day, Month, Year): ⌊⌊⌊⌊⌊⌊⌊⌊

Today's date (Day, Month, Year): 31 ⌊⌊⌊⌊⌊⌊⌊⌊

	Not at All	A Little	Quite a Bit	Very Much
1. Do you have any trouble doing strenuous activities, like carrying a heavy shopping bag or a suitcase?	1	2	3	4
2. Do you have any trouble taking a long walk?	1	2	3	4
3. Do you have any trouble taking a short walk outside of the house?	1	2	3	4
4. Do you need to stay in bed or a chair during the day?	1	2	3	4
5. Do you need help with eating, dressing, washing yourself or using the toilet?	1	2	3	4

During the past week:	Not at All	A Little	Quite a Bit	Very Much
6. Were you limited in doing either your work or other daily activities?	1	2	3	4
7. Were you limited in pursuing your hobbies or other leisure time activities?	1	2	3	4
8. Were you short of breath?	1	2	3	4
9. Have you had pain?	1	2	3	4
10. Did you need to rest?	1	2	3	4
11. Have you had trouble sleeping?	1	2	3	4
12. Have you felt weak?	1	2	3	4
13. Have you lacked appetite?	1	2	3	4
14. Have you felt nauseated?	1	2	3	4
15. Have you vomited?	1	2	3	4
16. Have you been constipated?	1	2	3	4

Please go on to the next page

SPECIMEN

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

During the past week:	Not at All	A Little	Quite a Bit	Very Much
17. Have you had diarrhea?	1	2	3	4
18. Were you tired?	1	2	3	4
19. Did pain interfere with your daily activities?	1	2	3	4
20. Have you had difficulty in concentrating on things, like reading a newspaper or watching television?	1	2	3	4
21. Did you feel tense?	1	2	3	4
22. Did you worry?	1	2	3	4
23. Did you feel irritable?	1	2	3	4
24. Did you feel depressed?	1	2	3	4
25. Have you had difficulty remembering things?	1	2	3	4
26. Has your physical condition or medical treatment interfered with your family life?	1	2	3	4
27. Has your physical condition or medical treatment interfered with your social activities?	1	2	3	4
28. Has your physical condition or medical treatment caused you financial difficulties?	1	2	3	4

For the following questions please circle the number between 1 and 7 that best applies to you

29. How would you rate your overall health during the past week?

1	2	3	4	5	6	7
Very poor						Excellent

30. How would you rate your overall quality of life during the past week?

1	2	3	4	5	6	7
Very poor						Excellent

SPECIMEN

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019

18.6. Description of the Clinical Trial in the AP-HP Trials Register

Les lymphomes T cutanés sont des cancers des globules blancs qui affectent principalement la peau. Les deux formes les plus courantes sont le mycosis fongoïde, qui se manifeste par des plaques ou tumeurs sur la peau, et le syndrome de Sézary, qui provoque une rougeur généralisée de la peau et la présence de cellules cancéreuses dans le sang.

Des chercheurs ont découvert qu'un récepteur, appelé CCR8, qui est une protéine présente à la surface des cellules cancéreuses, pourrait être une cible pour un nouveau traitement des lymphomes T cutanés. Un anticorps monoclonal, nommé DT-7012, a montré, dans des études en laboratoire qu'il peut tuer les cellules cancéreuses et renforcer les défenses naturelles du corps contre le cancer.

Ce traitement pourrait être une solution prometteuse pour les patients chez qui les traitements habituels ne fonctionnent pas ou plus.

Cette recherche porte donc sur l'évaluation de la tolérance de l'anticorps monoclonal anti-CCR8 (DT-7012) et la recherche de la dose la plus adaptée pour les patients atteints d'un lymphome cutané à cellules T récidivant ou réfractaire.

Pour répondre à la question posée, il est prévu d'inclure 30 patients présentant un lymphome cutané à cellules T récidivant ou réfractaire dans plusieurs établissements de soins situés en France, en Allemagne et en Espagne.
La durée prévisionnelle de la recherche est de 34 mois et la participation des patients sera de 13 mois, dont 12 mois de traitement maximum.

Cette recherche bénéficie d'un financement public français par l'ANR (Agence Nationale de la recherche).

This document is the property of DRCI/AP-HP. All reproduction is strictly prohibited.
Version 4.0 dated 31/05/2019